PROVEIT / RESEARCH CONTINUATION

Alternating Tails and Higher Spectral Defects

Cyclotomic generators, all-index Stieltjes identities,
Gamma primitives, and exact moment-reconstruction certificates

Prepared for Vladimir Reshetnikov's ProveIt programme
Mathematical derivations and computational audit with OpenAI assistance

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Abstract

We develop four exact continuations of the current polylogarithm manuscript and its incoming reports. A finite adjacent-tail certificate gives a complementary square identity; its continuous-order version produces ordinary convergent identities for every generalized Stieltjes index and an anchored Gamma primitive. A finite cyclotomic generator reduces the prescribed conjugation sector of rational-tail products at every conductor to trigonometric functions and digamma values, with explicit conductor-four evaluations. For a quadratic spectrum with an arbitrary polynomial weight, we compute every spectral derivative of the conjugate-factor defect. The answer separates an ordinary convergent logarithmic sum from finite Hurwitz and Stieltjes terms, including the pole contribution that is lost by regular-part differentiation. For any number of quadratic factors and every even polynomial weight, we determine the complete second derivative, a digamma double primitive of the mass correction, and all coincident-mass coefficients. Finally, exact Bernoulli congruences prove new infinite families of optimal moment prefixes at orders eight and ten, leaving one explicitly identified residue class at each order unresolved.

The proofs are analytic or exact algebraic. Symbolic certificates and independent high-precision diagnostics are supplied separately, with their different evidential roles stated. The article reconciles source status, preserves existing corrections, and proposes concrete further research questions. Novelty is relative to the inspected project corpus; global literature priority is not asserted.

Repository anchor. `VladimirReshetnikov/ProveIt`, commit `73f18ef013d5`.
Scope. Exact identities and proofs; no new formalization claim.
Companion artifacts. Modular and standalone LaTeX, exact certificates, verification programs, recorded results, and integration notes.

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1 Scope, conventions, and the results established

This article continues the exact-identity programme in the unified ProveIt manuscript and the six archives present in its incoming directory at commit

`73f18ef013d5b74eafc883efb2c15210642df12a`.

The commit is dated 11 October 2026, 18:22:01 UTC. The incoming archive interiors were retrieved and inspected for overlap with the selected targets. The repository manuscript and the relevant primary references were checked where the proofs use them. This is a targeted mathematical audit of those dependencies, not an exhaustive certification of every statement in the repository. The deliverable is a separate article; repository integration can preserve the existing thematic organization.

The main technical theme is to retain exact finite differences and finite asymptotic terms before taking a limit. For tails, the missing information is an endpoint product. For zeta continuation, it is a finite residue contribution. For moment reconstruction, it is an exact rational determinant rather than a floating rank test. These three mechanisms lead to complementary identities among the functions named in the research programme.

1.1 A map from source questions to the present results

Source target	Result established here	Precise status
Harmonic-tail square identities	A rational-tail finite certificate, continuous order, all-index Stieltjes products, and anchored Gamma integration (Section 2)	Additional identities with ordinary convergence.
Shifts and Twists, question Q6 [3]	A finite generator for arbitrary root-of-unity inner and outer colors (Theorem 3.1)	Solved for the specified parity/conjugation sector; the complementary sector remains open.
Quadratic-Spectrum Stieltjes, question 2 [5]	Every order Δ_r for every polynomial weight (Theorem 4.3)	Complete explicit correction formula; generic finite polylogarithm reduction is not claimed.
The same report, question 7	Any number of quadratic factors, every even polynomial weight, and every coincident-mass coefficient (Theorems 6.1–6.3)	Complete at the second spectral derivative in this setting.
Optimal moment prefix, order eight [6]	Full formal rank for $p = 8$, $q = 2b$, $b \geq 5$, outside $8001 \pmod{16384}$ (Theorem 7.2)	Strengthens the existing exclusion $1 \pmod{16}$; one smaller progression remains.
Optimal moment prefix, order ten	Full formal rank for $p = 10$, $q = 2b$, $b \geq 6$, outside $962 \pmod{2048}$ (Theorem 7.3)	New infinite family and exact finite base cases; no all-order theorem is inferred.

The gap-four moment-prefix theorem is already established in the two latest harmonic-tail reports [4, 3]. It is not reproved here as a new result. The Gaussian weight-six and revised

weight-eight numerical conjectures are not settled by this article. The new cyclotomic identities concern different, explicitly specified sums; their existence is not a missing proof of those numerical equalities.

1.2 Conventions fixed throughout

The Hurwitz zeta function is initially $\zeta(s,a)=\sum_{n\ge0}(n+a)^{-s}$ for $\Re s>1$, $\Re a>0$, then continued in its usual variables. Our generalized Stieltjes constants are defined by

$$\zeta(1+u,a)=\frac{1}{u}+\sum_{r\ge0}\frac{(-1)^r}{r!}\gamma_r(a)u^r. \tag{1.1}$$

Thus $\gamma_0(a)=-\psi(a)$, $\gamma_0(1)=\gamma$, and a literal linear Laurent coefficient is $-\gamma_1(a)$. The prime in $\zeta'(s,a)$ always differentiates s; a derivative in a is written ∂_a. We use

$$\psi(a)=\frac{\Gamma'(a)}{\Gamma(a)},\qquad \psi^{(r)}(a)=(-1)^{r+1}r!\zeta(r+1,a)\quad(r\ge1),$$

and $H_n^{(r)}=\sum_{k=1}^{n}k^{-r}$, with $H_n=H_n^{(1)}$ and $H_0=0$. The polylogarithm is $\mathrm{Li}_s(z)=\sum_{k\ge1}z^k/k^s$ in its convergence disk, with continuation or ordinary boundary values stated when used. These conventions agree with the standard references [12, 13, 14].

All complex powers use principal logarithms on the initial positive shift domains. Any later continuation or antiderivative specifies its domain and its anchor. A square of a complex tail means its algebraic square; a modulus square is written explicitly. A series called *ordinary* is a limit of its stated partial sums. A continued spectral value is never silently substituted for a divergent ordinary sum. "Locally normally convergent" means locally uniform absolute convergence, sufficient for holomorphic parameter differentiation.

The symbols used for colors in Section 3 are roots of unity; the positive real shift and mass variables in Sections 4–6 are local to those sections. Nielsen's alternating Hurwitz function $\mathcal{B}_s$ is distinguished from the Bernoulli polynomials B_j and the cyclotomic generating function B_ω. The moment-rank statements concern a formal coordinate space in which only the specified exact reduction relations have been imposed. They make no arithmetic independence assertion about actual zeta values.

1.3 What counts as proof in the companion package

The infinite-dimensional assertions below have written analytic proofs, including convergence, pole cancellation, and branch control. Exact symbolic certificates verify the finite algebra used in those proofs. The moment nonvanishing theorems combine a uniform Bernoulli congruence with exact polynomial identities and finitely many rational base cases. High-precision evaluations supply independent diagnostics for signs, factors, and endpoint conventions. They are not interval certificates, proofs of an infinite family, or proofs of period independence. No statement in this delivery is claimed to have been formalized in Lean or another proof assistant.

2 A complementary square law from one finite difference

The incoming alternating-outer resolvent theorem [3] evaluates $\sum_{n\ge0}(-1)^n(2n+1)E_n^\alpha(x)E_n^\beta(y)$ through alternating double-zeta parity. There is a complementary ordinary-outer square law, with a direct finite certificate. Its continuous-order extension yields ordinary convergent identities involving generalized Stieltjes constants. The finite-difference mechanism is elementary; the results below are additions relative to the inspected project sources, not claims of literature priority. Products of zeta tails are a classical subject; see Hoffman's multiple-zeta treatment [8].

2.1 The exact finite certificate and its compatible color

Let P be a polynomial of degree $d \geq 2$ with $P(k) \neq 0$ for every positive integer k. Let $0 < |\lambda| \leq 1$, $\lambda \neq 1$, and define

$$E_n = \sum_{k>n} \frac{\lambda^k}{P(k)}, \qquad t = -\lambda^{-1}, \qquad Q_n = t^n P(n+1) E_n E_{n+1} \quad (n \geq 0).$$

The value $P(0) = 0$ is allowed. All squares and products are algebraic; complex conjugation is not inserted.

Theorem 2.1 (A square identity for rational tails)**.** *The following series is absolutely convergent:*

$$\sum_{n\geq 0} t^n \{P(n+1) + \lambda P(n)\} E_n^2 = \lambda E_0 \{1 + P(0) E_0\}. \tag{2.1}$$

More precisely, for every integer $N \geq 0$,

$$\sum_{n=0}^{N} t^n \{P(n+1) + \lambda P(n)\} E_n^2 = \lambda E_0 \{1 + P(0) E_0\} + Q_N. \tag{2.2}$$

It holds locally normally in the coefficients of P on sets of fixed degree avoiding the positive-integer poles, for each fixed admissible λ.

Proof. For $n \geq 1$, substitute $E_{n-1} = E_n + \lambda^n / P(n)$ and $E_{n+1} = E_n - \lambda^{n+1}/P(n+1)$ into the difference of two successive Q's. Since $t^{-1} = -\lambda$, the linear-tail terms cancel:

$$Q_n - Q_{n-1} = t^n \{P(n+1) + \lambda P(n)\} E_n^2.$$

At the lower endpoint, $Q_0 = P(1) E_0^2 - \lambda E_0$, whereas the $n = 0$ summand is $\{P(1) + \lambda P(0)\} E_0^2$. Their difference is exactly the right side of (2.1). This proves the finite identity without evaluating $1/P(0)$.

When $|\lambda| < 1$, $E_n = O(|\lambda|^{n+1} n^{-d})$; the factor $|t|^n$ therefore leaves geometric decay. On the unit circle with $\lambda \neq 1$, summation by parts and the bounded partial sums of λ^k give $E_n = O(n^{-d})$: use $P(k)^{-1} = O(k^{-d})$ and $P(k+1)^{-1} - P(k)^{-1} = O(k^{-d-1})$. Thus the summands are $O(n^{-d})$, and $Q_N = O(N^{-d}) \to 0$. These estimates are uniform on compact coefficient sets as stated. □

The color constraint is forced within this short certificate class. If the same ansatz for Q_n is made with arbitrary $t \neq 0$, its difference is

$$t^n \left[\{P(n+1) - t^{-1} P(n)\} E_n^2 - \lambda^n (\lambda + t^{-1}) E_n \right].$$

Hence cancellation of the linear-tail term as an identity requires $t = -\lambda^{-1}$. This is a classification of the displayed ansatz, not an obstruction to other telescopers.

In particular, taking $P(k) = k^d$ gives an analytic polylogarithmic identity for every integer $d \geq 2$:

$$\sum_{n\geq 0} (-\lambda^{-1})^n \{(n+1)^d + \lambda n^d\} \left\{ \mathrm{Li}_d(\lambda) - \sum_{k=1}^{n} \frac{\lambda^k}{k^d} \right\}^2 = \lambda \, \mathrm{Li}_d(\lambda). \tag{2.3}$$

At $\lambda = 0$ both sides have a removable value zero; the tails contain λ^{n+1}, so no negative power survives in a summand. For example, $d = 2$ and $\lambda = i$ give the Gaussian identity

$$\sum_{n\geq 0} i^n \{(n+1)^2 + i n^2\} \left\{ \mathrm{Li}_2(i) - \sum_{k=1}^{n} \frac{i^k}{k^2} \right\}^2 = -G - \frac{i\pi^2}{48},$$

where $G = \beta(2)$ is Catalan's constant. This follows from $\mathrm{Li}_2(i) = -\pi^2/48 + iG$.

For $\lambda = -1$, $t = 1$ and the weight is $P(n+1) - P(n)$, whose degree is at most $d-1$. The same proof in this case permits $d = 1$; the summands are then $O(n^{-2})$.

At a root of unity of order N, the endpoint E_0 has a finite polygamma reduction. Indeed, partial fractions reduce $1/P(k)$ to terms $(k+z)^{-j}$, and residue classes give

$$\sum_{k=1}^{\infty} \frac{\lambda^k}{(k+z)^j} = N^{-j} \sum_{r=1}^{N} \lambda^r \zeta\left(j, \frac{r+z}{N}\right) \qquad (j \geq 2).$$

For $j = 1$ the correct pole-cancelled interpretation is

$$-\frac{1}{N} \sum_{r=1}^{N} \lambda^r \psi\left(\frac{r+z}{N}\right),$$

because $\sum_{r=1}^{N} \lambda^r = 0$. These formulas follow first for shifts in the right half-plane and then meromorphically. They show that (2.1) gives finite rational-color Gamma/polygamma identities without appealing to numerical period independence.

2.2 The resolvent square and a two-shift extension

Corollary 2.2 (A sine-square evaluation)**.** *For $x \notin \mathbb{Z} \setminus \{0\}$,*

$$\boxed{\sum_{n\geq 0} (2n+1) \left(\sum_{k>n} \frac{(-1)^k}{k^2 - x^2} \right)^2 = \frac{\pi^2}{4\sin^2(\pi x)} - \frac{1}{4x^2}.} \tag{2.4}$$

The apparent singularity at zero is removable, with value $\zeta(2)/2$.

Proof. Apply Theorem 2.1 to $P(k) = k^2 - x^2$ and $\lambda = -1$. The cotangent/cosecant partial fractions give

$$E_0 = \frac{1}{2x^2} - \frac{\pi}{2x\sin(\pi x)}.$$

The endpoint is $x^2 E_0^2 - E_0$, which is the displayed expression. The proof also establishes meromorphic continuation and local normal convergence away from the stated coordinate poles. □

For example, at $x = 1/2, 1/3, 1/4$ the values are, respectively,

$$\frac{\pi^2}{4} - 1, \qquad \frac{\pi^2}{3} - \frac{9}{4}, \qquad \frac{\pi^2}{2} - 4.$$

For real $y \neq 0$, substituting $x = iy$ instead gives

$$\sum_{n\geq 0} (2n+1) \left(\sum_{k>n} \frac{(-1)^k}{k^2 + y^2} \right)^2 = \frac{1}{4y^2} - \frac{\pi^2}{4\sinh^2(\pi y)}.$$

If $T_s^-(n) = \sum_{k>n} (-1)^k k^{-s}$, coefficient extraction in (2.4) gives, for every integer $m \geq 2$,

$$\sum_{a=1}^{m-1} \sum_{n\geq 0} (2n+1) T_{2a}^-(n) T_{2m-2a}^-(n) = \frac{2m-3}{2} \zeta(2m-2). \tag{2.5}$$

To check the coefficient, differentiate $\pi\cot(\pi x) = x^{-1} - 2\sum_{r\geq 1} \zeta(2r) x^{2r-1}$. The resolvent series is normally convergent for $|x| < 1$, including after any fixed coefficient extraction.

Write

$$\mathcal{B}(z)=\frac{1}{2}\left\{\psi\left(\frac{z+1}{2}\right)-\psi\left(\frac{z}{2}\right)\right\},\quad M_n(a,b)=\frac{\mathcal{B}(n+a+1)-\mathcal{B}(n+b+1)}{b-a}.$$

The quotient has its derivative meaning at $a=b$. Partial fractions and the alternating digamma series identify it with $\sum_{j\geq 0}(-1)^j/((j+n+a+1)(j+n+b+1))$. Taking $P(k)=(k+a)(k+b)$ gives the all-shift identity

$$\sum_{n\geq 0}(2n+a+b+1)M_n(a,b)^2=M_0(a,b)-abM_0(a,b)^2. \tag{2.6}$$

It is meromorphic in a,b, with the positive-lattice denominator poles excluded. In particular, $a=0,b=1$ yields

$$\sum_{n\geq 0}(2n+2)\left(\sum_{k>n}\frac{(-1)^k}{k(k+1)}\right)^2=2\log 2-1.$$

2.3 Continuous order: an ordinary Hurwitz identity

Define the alternating Hurwitz family

$$\mathcal{B}_s(z)=\sum_{j\geq 0}\frac{(-1)^j}{(j+z)^s}=2^{-s}\left\{\zeta\left(s,\frac{z}{2}\right)-\zeta\left(s,\frac{z+1}{2}\right)\right\}. \tag{2.7}$$

The Hurwitz difference is entire in s, since its residues at one cancel. On $\Re s>0,\Re z>0$ the ordinary alternating series and

$$\mathcal{B}_s(z)=\frac{1}{\Gamma(s)}\int_0^\infty\frac{u^{s-1}e^{-zu}}{1+e^{-u}}\,du$$

agree. At $s=1$, this is the classical Nielsen beta function [10] $\mathcal{B}$; it is not Euler's two-variable beta function.

Theorem 2.3 (Continuous-order square law)**.** *For $\Re s>0$, $\Re a>0$, using the principal powers,*

$$\sum_{n\geq 0}\{(n+a+1)^s-(n+a)^s\}\mathcal{B}_s(n+a+1)^2 = \mathcal{B}_s(a+1)-a^s\mathcal{B}_s(a+1)^2. \tag{2.8}$$

The series is absolutely and locally normally convergent in (s,a). The formula also holds at real $a=0$, with $0^s=0$ in this half-plane. In that case the right side is the Dirichlet eta function $\eta(s)=(1-2^{1-s})\zeta(s)$, with its removable value at one.

Proof. The recurrence $\mathcal{B}_s(z)+\mathcal{B}_s(z+1)=z^{-s}$ implies, for $F_s(z)=z^s\mathcal{B}_s(z)\mathcal{B}_s(z+1)$,

$$F_s(z-1)-F_s(z)=\{z^s-(z-1)^s\}\mathcal{B}_s(z)^2.$$

Summing this finite identity gives the stronger statement

$$\begin{aligned}&\sum_{n=0}^{N}\{(n+a+1)^s-(n+a)^s\}\mathcal{B}_s(n+a+1)^2\\&=\mathcal{B}_s(a+1)-a^s\mathcal{B}_s(a+1)^2-(N+a+1)^s\mathcal{B}_s(N+a+1)\mathcal{B}_s(N+a+2).\end{aligned} \tag{2.9}$$

The lower endpoint form follows from the same recurrence and does not require evaluating $\mathcal{B}_s(0)$ at $a=0$.

The Laplace representation, or paired summation by parts, gives $\mathcal{B}_s(n+a+1) = O(n^{-\Re s})$ locally uniformly. The weight difference is $O(n^{\Re s-1})$, since it equals $s\int_0^1 (n+a+v)^{s-1}\,dv$ for large n. The summands are therefore $O(n^{-\Re s-1})$, while the remaining endpoint in (2.9) is $O(N^{-\Re s})$. This proves the asserted convergence and permits every fixed spectral and shift derivative by Cauchy's formula. □

The integer specializations give

$$\sum_{n\ge 0}\{(n+1)^d - n^d\}\mathcal{B}_d(n+1)^2 = \eta(d) \qquad (d=1,2,\dots).$$

More generally, for $d \ge 2$, the integrands and summands in (2.8) are finite polygamma expressions because

$$\mathcal{B}_d(z) = \frac{(-1)^d}{2^d(d-1)!}\left\{\psi^{(d-1)}\left(\frac{z}{2}\right) - \psi^{(d-1)}\left(\frac{z+1}{2}\right)\right\}.$$

2.4 All Stieltjes orders with no finite-part prescription

Use the Laurent convention $\zeta(1+u,z) = u^{-1} + \sum_{j\ge 0}(-1)^j\gamma_j(z)u^j/j!$. Put $L = \log 2$ and define

$$C_r(n) = \sum_{j=0}^{r}\binom{r}{j}L^{r-j}\left\{\gamma_j\left(\frac{n+1}{2}\right) - \gamma_j\left(\frac{n+2}{2}\right)\right\},$$
$$W_k(n) = (n+1)\log^k(n+1) - n\log^k n.$$

At $n=0$, take $n\log^k n = 0$ and $\log^0(n+1) = 1$; thus $W_0(n) = 1$ for every n. Equation (2.7) immediately gives

$$\partial_s^r \mathcal{B}_s(n+1)|_{s=1} = \frac{(-1)^r}{2}C_r(n).$$

Corollary 2.4 (An all-index Stieltjes square identity)**.** *For each integer $r \ge 0$,*

$$\frac{1}{4}\sum_{n\ge 0}\sum_{k+\ell+m=r}\frac{r!}{k!\ell!m!}(-1)^k W_k(n)C_\ell(n)C_m(n)$$
$$= \frac{L^{r+1}}{r+1} - \sum_{j=1}^{r}\binom{r}{j}L^j\gamma_{r-j}. \tag{2.10}$$

Every fixed summand family is absolutely convergent. The empty right-hand sum at $r=0$ is zero.

Proof. Differentiate (2.8) r times in s at $a=0, s=1$. Normal convergence has been proved on a larger compact neighborhood of that point in the spectral half-plane. Leibniz's rule gives the left side, after multiplication by $(-1)^r$. Expanding $(1-e^{-Lu})\zeta(1+u)$ gives

$$\eta^{(r)}(1) = (-1)^r\left\{\frac{L^{r+1}}{r+1} - \sum_{j=1}^{r}\binom{r}{j}L^j\gamma_{r-j}\right\}.$$

This is the required right side. More explicitly, $C_j(n) = O(n^{-1}\log^j(n+1))$ and $W_k(n) = O(\log^k(n+1))$, so all the displayed scalar series converge absolutely. These estimates also follow by differentiating the paired alternating series, keeping its cancellation. □

For instance $\sum_{n\ge 0}C_0(n)^2 = 4\log 2$, and the first logarithmic identity is

$$\frac{1}{4}\sum_{n\ge 0}\{W_1(n)C_0(n)^2 - 2C_0(n)C_1(n)\} = \gamma\log 2 - \frac{1}{2}\log^2 2.$$

At second order the corresponding right side is $\frac{1}{3}\log^3 2 - \gamma\log^2 2 - 2\gamma_1\log 2$. These are ordinary convergent identities on the half-lattice, not distributional contact terms or finite parts.

2.5 An anchored Gamma primitive and shift derivatives

At $s = 1$, (2.8) becomes

$$\sum_{n\geq 0} \mathcal{B}(n+a+1)^2 = \mathcal{B}(a+1) - a\mathcal{B}(a+1)^2. \tag{2.11}$$

For real $u \geq 0$, its compensated primitive is

$$\begin{aligned}&\int_0^u \left\{ \sum_{n\geq 0} \mathcal{B}(n+a+1)^2 + a\mathcal{B}(a+1)^2 \right\} da \\ &= \log \frac{\sqrt{\pi}\,\Gamma(1+u/2)}{\Gamma((1+u)/2)}.\end{aligned} \tag{2.12}$$

Every exchange follows from local normal convergence, and the constant is fixed by $u = 0$. The same identity holds holomorphically on a simply connected pole-free neighborhood with the logarithm anchored at zero. At $u = 1$, the value is $\log(\pi/2)$.

Finally, because $\partial_a^j \mathcal{B}(n+a+1) = (-1)^j j! \mathcal{B}_{j+1}(n+a+1)$, every shift derivative of (2.11) gives an explicit mixed-order identity. Writing $z = a+1$ and $U_r(z) = \sum_{j=0}^{r} \mathcal{B}_{j+1}(z)\mathcal{B}_{r-j+1}(z)$, $U_{-1} = 0$, one obtains for $r \geq 0$

$$\sum_{n\geq 0} U_r(n+z) = \mathcal{B}_{r+1}(z) + U_{r-1}(z) - (z-1)U_r(z). \tag{2.13}$$

At positive integer orders the right side is a finite product of ordinary polygamma values at half-shifts. This is a convolution identity; it does not claim to isolate every individual mixed product.

Focused next questions. The diagonal finite certificate suggests a precise bilinear question: for distinct polynomial denominators P, Q, classify rational-coefficient certificates involving the two adjacent-tail products. A second target is meromorphic continuation of the continuous-order square law past $\Re s = 0$, with the explicit upper endpoint in (2.9) retained at each resonance. A third is to combine these matched-color identities with the independent two-parameter conjugation generator, determining which mixed resolvent coefficients are eliminated by their joint relation system. None of these questions is a numerical-independence claim.

3 A finite cyclotomic generator for products of rational tails

The incoming Shifts and Twists report [3] asks for an analogue of its two-parameter alternating identity at higher roots of unity. There is a precise answer for one conjugation sector, valid at every conductor. The answer below uses only elementary trigonometric functions and finitely many digamma values. Its coefficient form is consistent with the classical depth-two parity theorem for colored multiple polylogarithms [9]. We give an independent bilateral-sum proof, including the limiting conventions at the trivial color. No reduction of the opposite conjugation sector is asserted.

3.1 The finite functions

If ω is a root of unity, define, initially for $|t| < 1$,

$$F_\omega(t) = \begin{cases} -\psi(1-t) - \gamma, & \omega = 1, \\ -\dfrac{1}{N}\displaystyle\sum_{j=1}^{N} \omega^j \psi((j-t)/N), & \omega^N = 1, \quad \omega \neq 1. \end{cases} \tag{3.1}$$

The value is independent of the chosen multiple N of the conductor. For $\omega \neq 1$ it is the ordinary convergent series $\sum_{k\geq 1} \omega^k/(k-t)$; residue-class grouping and the digamma difference series prove (3.1). Its Taylor series is

$$F_\omega(t) = \sum_{j\geq 1} \mathrm{Li}_j(\omega) t^{j-1}.$$

For the trivial color the coefficient of t^0 in this last notation is set to zero; all coefficients of positive degree are their ordinary zeta values. This convention does not assign a value to a divergent harmonic series.

Write $\omega = e^{2\pi i\theta}$ with $0 < \theta < 1$ for a nontrivial color and put

$$B_\omega(t) = \begin{cases} \pi t \cot(\pi t), & \omega = 1, \\ \dfrac{\pi t e^{i\pi(2\theta-1)t}}{\sin(\pi t)}, & \omega \neq 1. \end{cases} \tag{3.2}$$

These functions have value 1 at zero. They obey

$$B_\omega(t) = 1 - tF_\omega(t) + tF_{\omega^{-1}}(-t). \tag{3.3}$$

For example $B_{-1}(t) = \pi t/\sin(\pi t)$. The coefficients $b_j(\omega) = [t^j]B_\omega(t)$ are explicit:

$$b_j(\omega) = \frac{(2\pi i)^j}{j!} B_j(\theta) \quad (\omega \neq 1), \qquad b_0(1) = 1, \quad b_1(1) = 0, \quad b_j(1) = -2\zeta(j) \ (j \geq 2 \text{ even}),$$

and $b_j(1) = 0$ for odd $j \geq 3$. The $B_j(\theta)$ here are ordinary Bernoulli polynomials, whereas $B_\omega(t)$ denotes their scaled generating function. Formula (3.3) also proves that for $j \geq 1$

$$b_j(\omega) = -\mathrm{Li}_j(\omega) - (-1)^j \mathrm{Li}_j(\omega^{-1}),$$

with the stated convention at $\omega = 1$, $j = 1$.

Define the following finite expression:

$$\begin{aligned} C_{a,b}(X,Y) = {} & \frac{B_{ab}(X)F_b(Y-X) - B_a(X)F_b(Y)}{X} \\ & - \frac{B_{ab}(Y)F_{a^{-1}}(Y-X) - F_{a^{-1}}(-X)}{Y}. \end{aligned} \tag{3.4}$$

Each numerator has the indicated factor, so $X = 0$ and $Y = 0$ are removable for this expression near the origin. We use the finite four-point operation

$$Q_{a,b}(x,y) = \frac{C_{a,b}(x,y) + C_{a,b}(x,-y) - C_{a,b}(-x,y) - C_{a,b}(-x,-y)}{4x}. \tag{3.5}$$

Finally set

$$J_\omega(t) = \frac{B_\omega(t) - B_\omega(-t)}{2t} = \begin{cases} 0, & \omega = 1, \\ i\pi \dfrac{\sin(\pi(2\theta-1)t)}{\sin(\pi t)}, & \omega \neq 1. \end{cases} \tag{3.6}$$

The function J_ω is even. Thus $[J_\omega(x) - J_\omega(y)]/(x^2 - y^2)$ has its usual derivative interpretation when $y = \pm x$.

3.2 All colors and an arbitrary outer character

For roots of unity a, b, c write

$$\begin{aligned} E_n^a(x) &= \sum_{k>n} \frac{a^k}{k^2 - x^2}, \\ R_{a,b;c}(x,y) &= \sum_{n\geq 0} c^n \{c(n+1) - n\} E_n^a(x) E_n^b(y). \end{aligned} \tag{3.7}$$

The normalization of the weight makes its partial sum exactly $c^m m$. In particular $R_{a,b;-1} = -\mathcal{K}_{a,b}$ for the alternating weight $\mathcal{K}_{a,b} = \sum(-1)^n(2n+1)E_n^a E_n^b$.

Theorem 3.1 (Cyclotomic rational-tail generator)**.** *Let a, b, c be arbitrary roots of unity and put $d = abc$. Initially assume $|x| + |y| < 1$ and $xy \neq 0$. Then*

$$\begin{aligned} &R_{a,b;c}(x,y) + R_{a^{-1},b^{-1};c^{-1}}(x,y) \\ &\qquad = Q_{a,bc}(x,y) + Q_{b,ac}(y,x) + \frac{J_d(x) - J_d(y)}{x^2 - y^2}. \end{aligned} \tag{3.8}$$

Both sides extend meromorphically to $\mathbb{C}^2$. All poles away from $x \in \mathbb{Z} \setminus \{0\}$ or $y \in \mathbb{Z} \setminus \{0\}$ are removable in the combined right side. The series is absolutely locally normally convergent unless $a = b = 1$ and $c \neq 1$. In that exceptional case it converges ordinarily and locally uniformly; subtracting the summand at $(x, y) = (0, 0)$ makes the remaining series absolutely locally normally convergent. When x, y are real, the left side of (3.8) is $2\Re R_{a,b;c}(x, y)$.

In particular this is a conductor-four generator, with quarter-shift digamma expressions furnished by (3.1). It also works for odd conductors, because the outer color c is arbitrary. The appearance of a conjugate pair is essential: the theorem proves a depth reduction for the real sector at odd total weight, and does not prove such a reduction for each complex summand separately.

For nonzero rational x, y with $x, y, x + y, x - y \notin \mathbb{Z}$, Gauss's rational digamma formula [15, Eq. (5.4.19)] makes the answer a finite explicit combination of π, π^2, and $\pi \log \alpha$ for positive algebraic α, with algebraic coefficients. The same assertion holds on $x = y \neq 0$ when $2x \notin \mathbb{Z}$. To see the stated coefficient class, F_ω at a rational argument is an algebraic linear combination of 1, π, and such logarithms: its Euler constant cancels in (3.1). Every B_ω value at a nonzero rational argument is π times an algebraic number. The apparently unmultiplied term $F_{a^{-1}}(-X)/Y$ in (3.4) is odd in Y and disappears in (3.5). The remaining J quotient is elementary, also at its diagonal limit. This statement concerns the displayed nonsingular rational parameters; the axes can introduce higher polylogarithmic constants, as the $\zeta(3)$ limits below demonstrate.

3.3 A bilateral proof with finite cutoff control

For nontrivial unit colors a, b, introduce the ordinary generating series

$$Z_{a,b}(X,Y) = \sum_{k>m\geq 1} \frac{a^k b^m}{(k-X)(m-Y)}.$$

Summation by parts in k proves locally uniform convergence: the inner partial sums grow at most logarithmically and their successive increments are $O(1/k)$. We claim

$$Z_{a,b}(X,Y) - Z_{a^{-1},b^{-1}}(-X,-Y) = C_{a,b}(X,Y) - \frac{F_{ab}(X) - F_{ab}(Y)}{X - Y}. \tag{3.9}$$

The last quotient is removable at $X = Y$.

To prove it, sum first over the full integer lattice in

$$S = \sum_{m\geq 1} b^m \sum_{k\in\mathbb{Z}} \frac{(ab)^k}{(k-X)(k+m-Y)}.$$

For each fixed m the inner series is absolutely convergent. The bilateral partial-fraction formula is

$$\mathrm{PV} \sum_{k\in\mathbb{Z}} \frac{\omega^k}{k-t} = -\frac{B_\omega(t)}{t}.$$

It follows either from (3.3) or from the elementary Fourier series of the shifted resolvent. The symmetric principal value at $\omega = 1$ is exactly the cotangent in (3.2). Partial fractions therefore give

$$S = -\frac{B_{ab}(X)}{X} F_b(Y - X) + \frac{B_{ab}(Y)}{Y} F_{a^{-1}}(Y - X). \tag{3.10}$$

Now put $l = k + m$. The positive, negative, mixed, and two zero-index parts of the region $l > k$ give respectively

$$\begin{aligned} S = {} & Z_{b,a}(Y, X) + Z_{a^{-1},b^{-1}}(-X, -Y) - F_{a^{-1}}(-X) F_b(Y) \\ & - F_b(Y)/X + F_{a^{-1}}(-X)/Y. \end{aligned}$$

This rearrangement has a direct cutoff justification. Keep $1 \le m \le M$ before separating regions. In each same-sign region, the omitted strip is bounded after summation by parts by a constant times

$$\sum_{k \ge 1} \frac{1}{k(k + M)} = O(\log M/M).$$

In the mixed region the difference between the triangular cutoff $u + v \le M$ and the product of the two ordinary sums is $O(\log M/M)$ as well: for $u < M$ use the nontrivial-color tail $O((M - u)^{-1})$, and for $u \ge M$ use the nontrivial-color tail in u. The two zero-index tails are $O(1/M)$. These estimates are uniform on compact parameter sets avoiding poles. Thus all the displayed limiting rearrangements are valid.

The elementary product decomposition is

$$Z_{b,a}(Y, X) = F_a(X) F_b(Y) - Z_{a,b}(X, Y) - \frac{F_{ab}(X) - F_{ab}(Y)}{X - Y}.$$

Use this identity, (3.3), and (3.10) to obtain (3.9).

Here are explicit conventions for the boundary colors. For $r \ge 2$, $s \ge 1$, let $w = r + s$ and write $L_j(\omega) = \mathrm{Li}_j(\omega)$. Coefficient extraction in (3.9) gives

$$\begin{aligned} & Z(r, s; a, b) - (-1)^w Z(r, s; a^{-1}, b^{-1}) \\ = {} & (-1)^s \sum_{j=s}^{w} (-1)^j \binom{j-1}{s-1} L_j(b) b_{w-j}(ab) - L_w(ab) \\ & + (-1)^r \sum_{j=r}^{w} \binom{j-1}{r-1} L_j(a^{-1}) b_{w-j}(ab) - L_s(b) b_r(a), \end{aligned} \tag{3.11}$$

where $Z(r, s; a, b) = \sum_{k > m \ge 1} a^k b^m / (k^r m^s)$. Every double series here is absolutely convergent. If $b = 1$, $s = 1$, the first summand with $j = 1$ cancels the last term before any value is assigned to $L_1(1)$. If $ab = 1$, the coefficient of $b_1(ab)$ vanishes, since $b = a^{-1}$ and $\binom{w-2}{s-1} = \binom{w-2}{r-1}$ with opposite signs. There is no other $L_1(1)$ term because $r \ge 2$. Consequently the finite formula, and its normally convergent even-r, odd-s generator, extend to every root-of-unity color with exactly (3.1)–(3.2). One may also obtain this extension by taking colors to 1 on the unit circle; the only discarded terms have $r = 1$, which the projection in (3.5) annihilates.

We now prove the tail theorem. A finite outer cutoff, followed by the telescoping identity

$$\sum_{n=0}^{m-1} c^n \{c(n + 1) - n\} = c^m m,$$

gives

$$R_{a,b;c}(x, y) = D_{a,bc}(x, y) + D_{b,ac}(y, x) + H_{abc}(x, y), \tag{3.12}$$

where

$$D_{u,v}(x,y)=\sum_{k>m\ge 1}\frac{u^k v^m m}{(k^2-x^2)(m^2-y^2)},$$
$$H_d(x,y)=\frac{O_d(x)-O_d(y)}{x^2-y^2},\qquad O_d(t)=\frac{F_d(t)+F_d(-t)}{2}.$$

The double sums are absolutely convergent, dominated by a constant times $\sum_{k>m}k^{-2}m^{-1}$. The omitted cutoff boundary is $O(M^{-1})$, even when both tails are untwisted. Applying the four-point operation to (3.9) gives

$$D_{u,v}+D_{u^{-1},v^{-1}}=Q_{u,v}-H_{uv}.$$

Add (3.12) to its inverse-color version. The two copies of $-H_d$, together with the two diagonal terms, leave $H_{d^{-1}}-H_d$. Formula (3.3) says $O_{d^{-1}}-O_d=J_d$, proving (3.8).

For completeness, locally uniformly away from the coordinate poles,

$$E_n^1(x)=O(n^{-1}),\qquad E_n^a(x)=O(n^{-2})\quad(a\neq 1),\qquad E_n^a(x)-E_n^a(0)=O(n^{-3}).$$

These imply every asserted absolute convergence statement. If $a=b=1$ and $c\neq 1$, the scalar summand at zero is $c^n\{(c-1)/n+O(n^{-2})\}$ for $n\ge 1$, which is ordinarily convergent; the parameter difference is $O(n^{-3})$. The case $c=1$ has weight 1 and is absolutely convergent even for $a=b=1$. Near a coordinate pole only finitely many tail terms are singular. Normal convergence and the meromorphic identity theorem therefore give the claimed continuation and all oblique removability.

3.4 Higher polynomial weights and their precise parity sector

For $p,q\ge 2$, $1\le h<\min(p,q)$, put

$$T_p^a(n)=\sum_{k>n}\frac{a^k}{k^p},\qquad M_{p,q,h}(a,b;c)=\sum_{n\ge 0}c^n\{c(n+1)^h-n^h\}T_p^a(n)T_q^b(n).$$

Exactly the same cutoff proof gives the ordinary identity

$$M_{p,q,h}(a,b;c)=Z(p,q-h;a,bc)+Z(q,p-h;b,ac)+L_{p+q-h}(abc). \tag{3.13}$$

The only possibly conditional case is $p=q=2$, $h=1$, $a=b=1$, $c\neq 1$; when $c=1$ the weight has one lower degree. Let $w=p+q-h$. Formula (3.11) now explicitly reduces

$$M_{p,q,h}(a,b;c)-(-1)^w M_{p,q,h}(a^{-1},b^{-1};c^{-1})$$

to finitely many single colored polylogarithm values and Bernoulli coefficients. For real parameters this is the real sector if w is odd and the imaginary sector if w is even. Thus the parity qualification is fully explicit at every conductor and every admissible weight.

3.5 Two conductor-four diagonal identities

Write

$$O_-(x)=-\log 2+O_+(x/2)-O_+(x),\qquad O_+(x)=-\frac{\psi(1+x)+\psi(1-x)+2\gamma}{2}.$$

For real x away from nonzero integers, define

$$D(x)=\Re\sum_{n\ge 0}(-1)^n(2n+1)E_n^i(x)^2,\qquad V(x)=\sum_{n\ge 0}(-1)^n(2n+1)|E_n^i(x)|^2.$$

Both are ordinary absolutely convergent sums. The all-color theorem gives the compact evaluations

$$D(x) = \frac{\pi}{2x\sin(\pi x)}\{\cos(\pi x/2)O_-(x/2) - \cos(\pi x)O_-(x)\}, \tag{3.14}$$

$$V(x) = \frac{\pi}{2x\sin(\pi x)}\{\cos(\pi x/2)O_-(x/2) - O_-(x)\}. \tag{3.15}$$

One verification of the simplification is to use

$$O_i(t) + O_{-i}(t) = O_-(t/2), \quad \frac{B_i(t) + B_i(-t)}{2} = \frac{\pi t\cos(\pi t/2)}{\sin(\pi t)}, \quad B_1(t) = \pi t\cot(\pi t)$$

in (3.4)–(3.5). At $a = b = i$ the product color in the ordered sum is 1, while at $a = i, b = -i$ it is -1; in either case J vanishes. The terms at $F_{\pm i}(0)$ cancel in the conjugate sum. This gives (3.14) and (3.15) directly, without a limiting singular quotient.

In particular, subtracting these two formulas proves the especially short sine-tail square

$$\boxed{\sum_{n\geq 0}(-1)^n(2n+1)\left(\sum_{k>n}\frac{\sin(\pi k/2)}{k^2-x^2}\right)^2 = -\frac{\pi}{4x}\tan(\pi x/2)O_-(x).} \tag{3.16}$$

The algebraic-square version extends meromorphically to complex x; no complex conjugation is inserted there. Its removable value at zero is $\pi^2\log 2/8$. Two rational values are

$$\sum_{n\geq 0}(-1)^n(2n+1)\left(\sum_{k>n}\frac{\sin(\pi k/2)}{k^2-1/4}\right)^2 = \frac{\pi}{2}, \tag{3.17}$$

$$\sum_{n\geq 0}(-1)^n(2n+1)\left(\sum_{k>n}\frac{\sin(\pi k/2)}{k^2-1/16}\right)^2 = \pi(\sqrt{2}-1)\{2-\sqrt{2}\log(1+\sqrt{2})\}. \tag{3.18}$$

Here $O_-(1/2) = -1$ and $O_-(1/4) = \sqrt{2}\log(1+\sqrt{2}) - 2$, by the rational digamma formula. The same substitution gives

$$D(1/2) = \pi\{\log(1+\sqrt{2}) - \sqrt{2}\}, \qquad V(1/2) = \pi\{1 + \log(1+\sqrt{2}) - \sqrt{2}\}.$$

The coefficient limits are

$$D(0) = \frac{9}{32}\zeta(3) - \frac{3\pi^2\log 2}{16}, \qquad V(0) = \frac{9}{32}\zeta(3) + \frac{\pi^2\log 2}{16}.$$

For the cosine tails, the corresponding square has value $9\zeta(3)/32 - \pi^2\log 2/16$ at zero. These are evaluations of an ordinary alternating outer sum, so a negative value of the cosine square causes no positivity issue.

3.6 An anchored quarter-color Gamma primitive

Let $S(x)$ denote the holomorphic function in (3.16) on $|x| < 1$. Then

$$\boxed{\int_0^u x\cot(\pi x/2)S(x)\,dx = \frac{\pi}{8}\log\left(\frac{\Gamma(1+u/2)\Gamma((1-u)/2)}{\Gamma(1-u/2)\Gamma((1+u)/2)}\right), \qquad |u| < 1.} \tag{3.19}$$

The integrand has a removable singularity at zero. The logarithm is its holomorphic branch equal to zero at $u = 0$, and the path can be any path in the unit disk. Indeed (3.16) makes the integrand $-\pi O_-(x)/4$. Integrating the displayed digamma formula for O_- and then applying Gamma duplication yields (3.19); the two sides vanish at zero. Absolute local normal convergence permits integration term by term in the defining sum for S. A concrete endpoint is

$$\int_0^{1/2} x\cot(\pi x/2)S(x)\,dx = \frac{\pi}{4}\log\frac{\Gamma(1/4)}{2\Gamma(3/4)}.$$

This primitive is a direct conductor-four continuation of the earlier alternating diagonal Gamma primitive, with its additive constant fixed explicitly.

4 Every spectral derivative of the conjugate-factor defect

The quadratic-spectrum report [5] proves a first-derivative factorization law and asks, in its second further-research question, for the higher-derivative correction. The answer below consists of an ordinary convergent logarithmic sum and finitely many explicitly stated Hurwitz or Stieltjes terms. It works for every polynomial weight and every positive mass. The finite term arising from one extra power of the spectral variable is essential.

4.1 Conventions and the classical first derivative

Let $a, b > 0$, put $x_n = n + a$ and $Q_b(x) = x^2 + b^2$, and let $W(x) = \sum_{d=0}^{D} w_d x^d \in \mathbb{C}[x]$. Initially in their half-planes of absolute convergence, define

$$F_{W,\pm}(s) = \sum_{n\geq 0} W(x_n)(x_n \pm ib)^{-s}, \qquad Z_W(s) = \sum_{n\geq 0} W(x_n)Q_b(x_n)^{-s}. \tag{4.1}$$

The powers use principal logarithms. Since $x_n > 0$, $\log(x_n + ib) + \log(x_n - ib) = \log Q_b(x_n)$ without a branch correction. All evaluations at zero refer to meromorphic continuation. We use the normalization requested in [5]:

$$\mathscr{D}_W(s) = \frac{F_{W,+}(2s) + F_{W,-}(2s)}{2} - Z_W(s), \tag{4.2}$$

$$\Delta_r(W; a, b) = \mathscr{D}_W^{(r)}(0) = 2^{r-1}\{F_{W,+}^{(r)}(0) + F_{W,-}^{(r)}(0)\} - Z_W^{(r)}(0). \tag{4.3}$$

Primes and parenthesized derivatives on these spectra always differentiate their order, not their shift or mass.

For reference, the established local anomaly is

$$\Delta_1(W; a, b) = \frac{1}{4}[x^{-1}]W(x)\left\{\log(1 + ib/x) - \log(1 - ib/x)\right\}^2 \tag{4.4}$$

$$= \sum_{k\geq 1} w_{2k-1}\frac{(-1)^k b^{2k}}{k}\left(H_{2k} - \frac{1}{2}H_k\right). \tag{4.5}$$

Only finitely many terms occur. Its independence of a and its vanishing for every b precisely when W is even are inherited results, not new claims of this article. The general local-anomaly framework is classical; see Dowker [11]. We recover the formula below to fix the normalization and expose the additional terms at higher orders.

4.2 Finite harmonic coefficients and global subtraction

For $k \geq 1$ define the polynomial

$$C_k(s; b) = (-1)^k b^{2k}\left\{\frac{(2s)_{2k}}{(2k)!} - \frac{(s)_k}{k!}\right\} = \sum_{m=2}^{2k} \alpha_{k,m}(b)s^m. \tag{4.6}$$

Here $(s)_k = s(s+1)\cdots(s+k-1)$. For a finite list v, $e_j(v)$ denotes its elementary symmetric polynomial, with $e_0 = 1$ and $e_j = 0$ beyond the length of the list. Set

$$E_j(N) = e_j(1, 1/2, \ldots, 1/N), \qquad E_j(0) = \begin{cases} 1 & j = 0, \\ 0 & j > 0. \end{cases} \tag{4.7}$$

These are polynomials in the generalized harmonic numbers:

$$\sum_{j\geq 0} E_j(N)t^j = \exp\left(\sum_{\ell\geq 1}\frac{(-1)^{\ell-1}}{\ell}H_N^{(\ell)}t^\ell\right).$$

Lemma 4.1 (All local coefficients)**.** *For $m \geq 2$,*

$$\alpha_{k,m}(b) = \frac{(-1)^k b^{2k}}{k}\{2^{m-1}E_{m-1}(2k-1) - E_{m-1}(k-1)\}. \tag{4.8}$$

Set $\alpha_{k,m} = 0$ for $m < 2$ or $m > 2k$. In particular,

$$C_1(s;b) = -2b^2s^2, \qquad \alpha_{k,2} = \frac{(-1)^k b^{2k}}{k}(2H_{2k-1} - H_{k-1}).$$

Proof. Use

$$\frac{(s)_k}{k!} = \frac{s}{k}\prod_{j=1}^{k-1}(1+s/j), \qquad \frac{(2s)_{2k}}{(2k)!} = \frac{s}{k}\prod_{j=1}^{2k-1}(1+2s/j).$$

Their linear terms agree. Comparing every subsequent coefficient proves the assertion, including the stated finite support. □

Put

$$h(s,x) = \frac{(x+ib)^{-2s} + (x-ib)^{-2s}}{2} - Q_b(x)^{-s}, \qquad K = \left\lfloor \frac{D+1}{2} \right\rfloor. \tag{4.9}$$

For $D = 0$ the integer K is zero and all sums from 1 to K are empty.

Lemma 4.2 (A globally valid analytic subtraction)**.** *In a complex neighborhood of $s = 0$,*

$$\begin{aligned}\mathscr{D}_W(s) = &\sum_{n\geq 0} W(x_n)\left\{h(s,x_n) - x_n^{-2s}\sum_{k=1}^{K} C_k(s;b)x_n^{-2k}\right\} \\ &+ \sum_{d=0}^{D}\sum_{k=1}^{K} w_d C_k(s;b)\zeta(2s+2k-d,a).\end{aligned} \tag{4.10}$$

The infinite sum converges locally normally. All the apparent singularities at zero in the finite sum are removable. The original three spectra in (4.1) are holomorphic at zero.

Proof. For $x > b$, the ordinary binomial expansions give

$$h(s,x) = x^{-2s}\sum_{k\geq 1} C_k(s;b)x^{-2k}.$$

On compact mass sets, its remainder after K terms is uniformly $O(x^{-2\Re s-2K-2})$ on a small closed s-disk. After multiplication by W, this is $O(x^{-2+2\varepsilon})$ when D is even and $O(x^{-3+2\varepsilon})$ when D is odd, for $|s| \leq \varepsilon < 1/4$. This proves normal convergence. The finitely many $x \leq b$ use the exact expression h, so no assumption $b < a$ is needed.

In a common right half-plane every term being subtracted is separately summable; adding it back as a Hurwitz zeta proves (4.10) there and hence by continuation. The only possible singular terms near zero have $2k - d = 1$. The simple pole of $\zeta(1+2s,a)$ is multiplied by $C_k = O(s^2)$, so they are removable. Finally, expansion of the polynomial weight about $x \pm ib$ expresses each $F_{W,\pm}$ as a finite sum of $\zeta(s-j, a \pm ib)$ with $j \geq 0$. They are holomorphic at zero, and the same follows for Z_W from (4.2). □

4.3 The complete higher-derivative formula

Define the elementary logarithmic polynomial

$$H_r(x;b) = (-1)^r\left[2^{r-1}\{\log^r(x+ib) + \log^r(x-ib)\} - \log^r Q_b(x)\right] \tag{4.11}$$

$$= \sum_{j=1}^{\lfloor r/2\rfloor}(-1)^{r+j}4^j\binom{r}{2j}\log^{r-2j}Q_b(x)\arctan^{2j}(b/x), \tag{4.12}$$

so $H_1 = 0$, $H_2 = -4\arctan^2(b/x)$, and $H_3 = 12\log Q_b(x)\arctan^2(b/x)$. Also put

$$P_{k,r}(L) = r! \sum_{m=2}^{\min(r,2k)} \alpha_{k,m}(b)\frac{(-2L)^{r-m}}{(r-m)!}. \tag{4.13}$$

Theorem 4.3 (Every order, every polynomial weight, every positive mass)**.** *For every integer* $r \geq 1$,

$$\Delta_r(W;a,b) = \sum_{n\geq 0} W(x_n)\left\{H_r(x_n;b) - \sum_{k=1}^{K}\frac{P_{k,r}(\log x_n)}{x_n^{2k}}\right\} + \sum_{d=0}^{D}\sum_{k=1}^{K} w_d \mathcal{J}_{k,r}(2k-d;a,b), \tag{4.14}$$

where, if $\nu \neq 1$,

$$\mathcal{J}_{k,r}(\nu;a,b) = r! \sum_{m=2}^{\min(r,2k)} \alpha_{k,m}(b)\frac{2^{r-m}}{(r-m)!}\zeta^{(r-m)}(\nu,a), \tag{4.15}$$

and at the resonant value $\nu = 1$,

$$\boxed{\mathcal{J}_{k,r}(1;a,b) = r!\left\{\frac{\alpha_{k,r+1}(b)}{2} + \sum_{m=2}^{\min(r,2k)} \alpha_{k,m}(b)\frac{(-2)^{r-m}}{(r-m)!}\gamma_{r-m}(a)\right\}.} \tag{4.16}$$

The infinite sum in (4.14) *is ordinary and absolutely convergent. It and every fixed parameter derivative converge locally normally on positive parameter sets.*

Proof. Differentiate the normally convergent expression (4.10). The derivative of its exact pointwise term is H_r; writing $\log(x \pm ib) = \frac{1}{2}\log Q_b(x) \pm i\arctan(b/x)$ proves (4.12). Differentiation of $x^{-2s}C_k(s;b)$ gives (4.13). If $\nu \neq 1$, Taylor multiplication gives (4.15).

At the pole, use the stipulated Laurent convention:

$$\zeta(1+2s,a) = \frac{1}{2s} + \sum_{j\geq 0}\frac{(-2)^j}{j!}\gamma_j(a)s^j.$$

The coefficient of s^r in $C_k(s;b)\zeta(1+2s,a)$ contains $\alpha_{k,r+1}/2$ from the pole and the displayed finite convolution from its regular part. Multiplication by $r!$ proves (4.16). The uniform estimates in the lemma allow the differentiations; logarithmic factors from spectral derivatives do not affect summability. The same estimates on small complex neighborhoods of positive (a,b) justify parameter derivatives. □

At $r = 1$ the ordinary sum and every nonresonant term vanish. The remaining term is $\alpha_{k,2}/2$, which recovers (4.5). At $r \geq 2$, deleting the ordinary sum is generally incorrect. Deleting $\alpha_{k,r+1}/2$ is also incorrect when this coefficient is present: differentiating a regular part alone loses the contribution of the pole.

4.4 A single monomial isolates every Stieltjes index

The simplest odd weight makes the correction especially transparent. For $r \geq 2$ set

$$c_r = (-1)^{r-1}2^{r-1}r(r-1).$$

Since $C_1 = -2b^2s^2$, the general theorem gives

$$\boxed{\Delta_r(x;a,b) = \sum_{n\geq 0}\left\{x_n H_r(x_n;b) - c_r b^2\frac{\log^{r-2}x_n}{x_n}\right\} + c_r b^2\gamma_{r-2}(a).} \tag{4.17}$$

Its residual summands are $O(x_n^{-3}(1+\log^{r-2} x_n))$. Thus this is an ordinary convergent identity for each generalized Stieltjes constant, coupled to precisely specified spectral derivatives. For example,

$$\Delta_2(x;a,b) = -4\sum_{n\geq 0}\left\{x_n \arctan^2(b/x_n) - \frac{b^2}{x_n}\right\} + 4b^2\psi(a), \tag{4.18}$$

$$\Delta_3(x;a,b) = 12\sum_{n\geq 0}\left\{x_n \log Q_b(x_n)\arctan^2(b/x_n) - \frac{2b^2\log x_n}{x_n}\right\} + 24b^2\gamma_1(a). \tag{4.19}$$

At first order, $\Delta_1(x;a,b) = -b^2$ even though each ordinary pointwise first-derivative difference is zero. This is a useful exact check on the chosen continuation, not a license to omit higher-order ordinary terms.

4.5 The first-derivative parity criterion fails completely at order two

Theorem 4.4 (Universal second-defect vanishing)**.** *Fix $a > 0$. For a polynomial $W \in \mathbb{C}[x]$,*

$$\Delta_2(W;a,b) = 0 \text{ for every } b > 0 \quad\Longleftrightarrow\quad W = 0.$$

Thus no nonzero even polynomial retains the universal vanishing property of the first derivative.

Proof. For $0 < b < a$, the full binomial expansion converges normally. If $k > (D+1)/2$, all the Hurwitz values $\zeta(2k-d,a)$ in its b^{2k} coefficient are ordinary convergent sums. That coefficient in Δ_2 is

$$\frac{2(-1)^k}{k}(2H_{2k-1} - H_{k-1})\sum_{n\geq 0}\frac{W(n+a)}{(n+a)^{2k}}. \tag{4.20}$$

The scalar is nonzero. If the defect vanishes identically, all these moments vanish. If $W \neq 0$, let n_0 be the least nonnegative integer with $W(n_0+a) \neq 0$. Multiplying the vanishing moment by $(n_0+a)^{2k}$ and letting $k \to \infty$ gives $W(n_0+a) = 0$, a contradiction. Dominated convergence is valid after fixing any sufficiently large k_0: polynomial growth of W makes $\sum_{n>n_0}|W(n+a)|((n_0+a)/(n+a))^{2k_0}$ finite. The reverse implication is immediate. □

This is a statement about identity in the mass at a fixed shift. It does not exclude accidental zeros at a prescribed mass for signed polynomial weights. No inequality or numerical independence assertion is needed for the proof.

5 The logarithmic correction as a Gamma integral and its primitives

For the constant weight no subtraction is necessary at any positive spectral derivative order. The second derivative produces a particularly simple ordinary function,

$$S(a,b) = \sum_{n\geq 0}\arctan^2\left(\frac{b}{n+a}\right), \qquad a,b > 0. \tag{5.1}$$

Theorem 4.3 immediately gives

$$Z_1''(0;a,b) = 2\zeta''(0,a+ib) + 2\zeta''(0,a-ib) + 4S(a,b). \tag{5.2}$$

The prime on ζ denotes its order derivative. Equation (5.2) is an exact higher-derivative replacement for a factorization rule, not a claim that the extra function is algebraically independent of some larger special-function class.

Theorem 5.1 (An ordinary logarithmic Gamma integral)**.** *For $a, b > 0$,*

$$\boxed{S(a,b) = 2\int_0^1 \frac{\log|\Gamma(a+ibt)| - \log|\Gamma(a+ib)|}{1-t^2}\,dt.} \tag{5.3}$$

The integrand has a removable value at $t = 1$. Both sides are real analytic in the positive parameters.

Proof. The Weierstrass product for the Gamma function, or the integrated digamma series, gives the ordinary normally convergent identity

$$\log|\Gamma(a+ibt)| - \log|\Gamma(a+ib)| = \frac{1}{2}\sum_{n\geq 0} \log\frac{(n+a)^2+b^2}{(n+a)^2+b^2t^2}.$$

For $0 \leq t \leq 1$ its summands are nonnegative, so Tonelli's theorem permits the required integral. The elementary identity

$$\int_0^1 \frac{\log((1+y^2)/(1+y^2t^2))}{1-t^2}\,dt = \arctan^2 y \qquad (y \geq 0) \tag{5.4}$$

finishes the proof. To verify it, differentiate in y; cancellation of $1-t^2$ leaves $2y(1+y^2)^{-1}\int_0^1(1+y^2t^2)^{-1}dt = 2\arctan y/(1+y^2)$, and both sides vanish at zero. The endpoint limit of the full integration density in (5.3), including the prefactor 2, is $b\,\Im\psi(a+ib)$. The defining series and its differentiated tails converge uniformly on compact positive parameter sets. □

Gamma reflection gives two elementary-kernel specializations:

$$S(1,b) = \int_0^1 \frac{\log\{t\sinh(\pi b)/\sinh(\pi bt)\}}{1-t^2}\,dt, \tag{5.5}$$

$$S(1/2,b) = \int_0^1 \frac{\log\{\cosh(\pi b)/\cosh(\pi bt)\}}{1-t^2}\,dt. \tag{5.6}$$

The endpoint at $t = 0$ in the first line is understood by continuity. We do not assert an unproved finite classical-polylogarithm evaluation of these integrals at generic b.

The small-mass expansion is an exact harmonic–Hurwitz generating function:

$$S(a,b) = \sum_{k\geq 1} \frac{(-1)^{k+1}b^{2k}}{k}\left(H_{2k} - \frac{1}{2}H_k\right)\zeta(2k,a), \qquad |b| < a. \tag{5.7}$$

It follows by expanding $\arctan^2 y$, or from (4.20). For every positive b, remove a finite head with $N + a > b$ and replace $\zeta(2k,a)$ in the tail by $\zeta(2k,N+a)$. This gives a convergent implementation without extending the disk of (5.7) improperly.

Two useful exact transport rules are

$$S(a,b) - S(a+1,b) = \arctan^2(b/a), \tag{5.8}$$

$$\sum_{j=0}^{q-1} S\left(\frac{a+j}{q}, \frac{b}{q}\right) = S(a,b) \qquad (q \in \mathbb{N}). \tag{5.9}$$

They follow by taking out the first term and by partitioning the positive shifted lattice into residue classes. The Gamma-integral formula gives the same distribution law through Gauss multiplication.

5.1 Every anchored shift antiderivative

Let $H_0 = 0$ and, for $m \geq 0$, define

$$J_m(z) = \frac{\zeta'(-m, z) + H_m \zeta(-m, z)}{m!}, \qquad \Re z > 0. \tag{5.10}$$

The combination is chosen to remove the extra term generated by differentiating the Pochhammer symbol.

Lemma 5.2. *For $m \geq 1$,*

$$\frac{d^m}{dz^m} J_m(z) = \zeta'(0, z) = \log \Gamma(z) - \frac{1}{2} \log(2\pi).$$

Proof. Differentiate the classical identity $\partial_z^m \zeta(s, z) = (-1)^m (s)_m \zeta(s + m, z)$ in s and set $s = -m$. Here $(s)_m = (-1)^m m!$ and its logarithmic derivative is $-H_m$. Thus $\partial_z^m \zeta'(-m, z) = m!\{\zeta'(0, z) - H_m \zeta(0, z)\}$, while $\partial_z^m \zeta(-m, z) = m! \zeta(0, z)$. The terms cancel as claimed. The final equality is the standard Hurwitz–Gamma formula [12]. □

Define

$$I_m(a, b) = 2\Re \int_0^1 \frac{J_m(a + ibt) - J_m(a + ib)}{1 - t^2}\, dt. \tag{5.11}$$

Then $I_0 = S$ and $\partial_a^m I_m = S$ for $m \geq 1$. For a prescribed base point $a_0 > 0$ the fully anchored primitive is

$$\begin{aligned} \widehat{I}_m(a, b; a_0) &= I_m(a, b) - \sum_{j=0}^{m-1} \frac{(a - a_0)^j}{j!} \partial_a^j I_m(a_0, b) \\ &= \frac{1}{(m-1)!} \int_{a_0}^{a} (a - u)^{m-1} S(u, b)\, du. \end{aligned} \tag{5.12}$$

All derivatives of orders below m vanish at $a = a_0$. The numerator in (5.11) is $O(1 - t)$ uniformly on compact parameter sets, so the displayed integrals and differentiations require no finite-part interpretation. Together with (5.2), this supplies explicit Hurwitz–Gamma primitives of the nonlocal second-derivative correction.

6 Several quadratic factors: the complete second derivative and confluence

The seventh further-research question in [5] asks for the second derivative of a spectrum with several masses and its coincident limit. The even-polynomial setting of that report admits a complete answer. In contrast to the first derivative, a logarithmic cross term remains; every coefficient of its coincident-mass expansion nevertheless has a finite polygamma and Bernoulli evaluation.

6.1 A finite definition for the weighted logarithmic kernel

Fix $a > 0$ and $P(v) = \sum_{d=0}^{D} p_d v^d$. For $B, C > 0$ set

$$\beta_k(B, C) = (-1)^k \sum_{j=1}^{k-1} \frac{(B^j - C^j)(B^{k-j} - C^{k-j})}{j(k - j)}, \qquad k \geq 2. \tag{6.1}$$

These are the coefficients in

$$\log^2 \frac{1 + Bz}{1 + Cz} = \sum_{k \geq 2} \beta_k(B, C) z^k.$$

They are symmetric in B, C and vanish when $B = C$. Define the following specific compensated kernel, with $x_n = n + a$:

$$K_{P,a}(B,C) = \sum_{n\geq 0} P(x_n^2)\left\{\log^2 \frac{x_n^2+B}{x_n^2+C} - \sum_{k=2}^{D} \frac{\beta_k(B,C)}{x_n^{2k}}\right\} + \sum_{d=0}^{D}\sum_{k=2}^{D} p_d \beta_k(B,C)\zeta(2k-2d,a). \tag{6.2}$$

All inner sums are empty when $D < 2$. The infinite sum has $O(x_n^{-2})$ tails and is ordinary and locally normally convergent. The finite counterterms have no pole: their Hurwitz orders are even, hence never one. For $P = 1$ or $P(v) = v$, the unmodified logarithmic sum already converges absolutely.

Theorem 6.1 (All even weights and any number of masses)**.** *For $q \geq 1$ and $\mathbf{B} = (B_1, \ldots, B_q)$ with each $B_j > 0$, let*

$$\mathcal{Z}_{P,\mathbf{B}}(s;a) = \sum_{n\geq 0} P(x_n^2) \prod_{j=1}^{q} (x_n^2 + B_j)^{-s}, \tag{6.3}$$

$$Z_{P,j}(s;a) = \sum_{n\geq 0} P(x_n^2)(x_n^2 + B_j)^{-s}. \tag{6.4}$$

Their continuations are holomorphic at zero, and

$$\boxed{\mathcal{Z}''_{P,\mathbf{B}}(0;a) = q\sum_{j=1}^{q} Z''_{P,j}(0;a) - \sum_{1\leq i<j\leq q} K_{P,a}(B_i,B_j).} \tag{6.5}$$

Repeated masses are allowed. Each coincident pair contributes zero.

Proof. Compare the product spectrum with $q^{-1}\sum_j Z_{P,j}(qs;a)$. At each lattice point write $L_j = \log(x^2 + B_j)$. The constant and first-order coefficients of

$$\frac{1}{q}\sum_j e^{-qsL_j} - e^{-s\sum_j L_j}$$

vanish. Its second derivative at zero is the elementary variance identity

$$q\sum_j L_j^2 - \left(\sum_j L_j\right)^2 = \sum_{i<j}(L_i - L_j)^2. \tag{6.6}$$

To justify this calculation after continuation, factor out x^{-2qs} and expand the remaining factors at infinity. Only even powers of x^{-1} occur, including after multiplication by $P(x^2)$. A pole at $s = 0$ would require a resulting x^{-1} term; there is none. Subtract finitely many asymptotic terms to leave a normally summable remainder. The constant and linear coefficients of each difference term vanish before the Hurwitz factors are differentiated. Hence its second derivative is exactly the finite-subtraction continuation of (6.6). The latter is (6.2) for each pair, proving (6.5). The same finite-subtraction argument proves holomorphy at zero for the original spectra. Repeated parameters follow either directly or from local normal convergence. □

For two unweighted factors, this reads

$$\mathcal{Z}''_{1,(B,C)}(0;a) = 2Z''_{1,B}(0;a) + 2Z''_{1,C}(0;a) - \sum_{n\geq 0} \log^2 \frac{x_n^2+B}{x_n^2+C}.$$

At $B = C$ it reduces to $4Z''_{1,B}(0;a)$, exactly as required by $\mathcal{Z}_{1,(B,B)}(s;a) = Z_{1,B}(2s;a)$.

6.2 An explicit digamma kernel determines the whole correction

The elementary quadratic resolvent is

$$R_a(B) = \sum_{n\geq 0} \frac{1}{x_n^2 + B} = \frac{\psi(a + i\sqrt{B}) - \psi(a - i\sqrt{B})}{2i\sqrt{B}}. \tag{6.7}$$

For a polynomial weight define its finite continuation

$$R_{P,a}(B) = P(-B)R_a(B) + \sum_{j=1}^{D} p_j \sum_{\ell=0}^{j-1} (-B)^\ell \zeta(-2(j-1-\ell), a). \tag{6.8}$$

It is the continued value $Z_{P,B}(1; a)$. Formula (6.8) follows from polynomial division of $x^{2j}/(x^2+B)$ and the Hurwitz values at nonpositive even integers. Those values are Bernoulli polynomials: $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$ [12].

Theorem 6.2 (A normalized double primitive of a digamma divided difference)**.** *For every P and positive B, C,*

$$\boxed{K_{P,a}(B, C) = \int_C^B \int_C^B \frac{R_{P,a}(u) - R_{P,a}(v)}{v - u}\, du\, dv.} \tag{6.9}$$

The integrand at $u = v$ is $-R'_{P,a}(u)$. In particular,

$$\partial_B \partial_C K_{P,a}(B, C) = -2\frac{R_{P,a}(B) - R_{P,a}(C)}{C - B}. \tag{6.10}$$

The kernel and both of its first partial derivatives vanish on the diagonal $B = C$.

Proof. At an ordinary lattice point,

$$\log^2 \frac{x^2 + B}{x^2 + C} = \int_C^B \int_C^B \frac{du\, dv}{(x^2 + u)(x^2 + v)}.$$

The rational kernel splits as

$$\frac{1}{(x^2 + u)(x^2 + v)} = \frac{(x^2 + u)^{-1} - (x^2 + v)^{-1}}{v - u}.$$

For an arbitrary polynomial weight, perform the same finite subtractions as in (6.2). Uniformity on the compact mass interval permits both integrals to pass through the ordinary remainder; the finitely many polynomial terms integrate directly. The resulting continued rational sum is the divided difference of (6.8), proving (6.9). The diagonal value is removable by differentiation. Differentiating the original logarithmic kernel and its finite subtractions gives (6.10). The diagonal normalizations follow from the square or from the double integral. □

6.3 Every coincident-mass coefficient is finite

Write $Z_P(m; B, a)$ for the continued value of $\sum_n P(x_n^2)(x_n^2 + B)^{-m}$. It is regular at every positive integer m, since possible spectral poles for an even polynomial weight occur at half-integers.

Theorem 6.3 (The full confluent expansion)**.** *For $|h| < a^2 + B$,*

$$\boxed{K_{P,a}(B, B + h) = 2 \sum_{m\geq 2} \frac{(-1)^m H_{m-1}}{m}\, h^m Z_P(m; B, a).} \tag{6.11}$$

Every coefficient has the finite evaluation

$$Z_P(m;B,a)=\frac{(-1)^{m-1}}{(m-1)!}\partial_B^{m-1}R_{P,a}(B). \tag{6.12}$$

Consequently no new unspecified constants are introduced in this coincident-mass expansion: the coefficients belong to the finite algebra generated by polygamma values at $a\pm i\sqrt{B}$, Bernoulli-polynomial values, and rational functions of B.

Proof. Use the ordinary Taylor identity, with the logarithm that vanishes at zero,

$$\log^2(1+z)=2\sum_{m\geq 2}\frac{(-1)^m H_{m-1}}{m}z^m, \qquad |z|<1.$$

The nearest possible zero of x_n^2+B+h occurs at $h=-(a^2+B)$. On every smaller disk the finite-subtracted logarithmic sum is holomorphic and the same subtractions justify coefficient extraction. More explicitly, separate the finitely many indices $m\leq D$, which use continued values, from $m\geq D+1$, whose weighted sums converge ordinarily and normally on the smaller disk. A coefficient is precisely the continued weighted resolvent $Z_P(m;B,a)$, which proves (6.11). Finally, differentiate the identity $\partial_B Z_P(s;B,a)=-sZ_P(s+1;B,a)$, first in its convergence region and then by continuation, starting at $s=1$. No integer-order pole is crossed. This gives (6.12) and the asserted finite special-function description. □

Equivalently, polynomial Taylor expansion about $-B$ gives

$$Z_P(m;B,a)=\sum_{\ell=0}^{D}\frac{P^{(\ell)}(-B)}{\ell!}Z_1(m-\ell;B,a).$$

For $j\geq 0$ the required nonpositive-order values are simply

$$Z_1(-j;B,a)=\sum_{\ell=0}^{j}\binom{j}{\ell}B^{j-\ell}\zeta(-2\ell,a),$$

and the positive orders are derivatives of (6.7). This second formula is convenient for exact symbolic implementation.

For instance, when $P=1$,

$$K_{1,a}(B,B+h)=h^2Z_1(2;B,a)-h^3Z_1(3;B,a)+\frac{11}{12}h^4Z_1(4;B,a)+O(h^5).$$

At $a=1/2$, $B=b^2$, the first coefficient is elementary:

$$Z_1(2;b^2,1/2)=\frac{\pi\tanh(\pi b)}{4b^3}-\frac{\pi^2\operatorname{sech}^2(\pi b)}{4b^2}. \tag{6.13}$$

It follows by differentiating $R_{1/2}(b^2)=\pi\tanh(\pi b)/(2b)$. The corresponding bilateral results follow by adding the formulas at a and $1-a$, for $0<a<1$. In that sum all nonpositive even Hurwitz values cancel by Bernoulli reflection, leaving an even shorter weighted resolvent. The higher spectral derivatives at a generic noncoincident mass remain described by the exact logarithmic kernels; no unproved finite polylogarithm reduction is inferred from the finite confluent coefficients.

7 Higher precision and the order-ten optimal prefix

The current reconstruction report proves the order-eight prefix outside $b\equiv 1 \pmod{16}$. We retain its exact four-square contraction and its integral numerator P_8; that reduction is inherited from [6]. The new contribution is a higher precision congruence that treats most of that exceptional class, together with a new five-square contraction and nonvanishing theorem at order ten. No assertion below settles the entire gap-six family or all unequal even orders.

7.1 Notation and the source reduction

For an integer $s \geq 2$, let

$$A_s(n) = \zeta(s) - H_n^{(s)} = \zeta(s, n+1), \qquad x_n = n + \tfrac{1}{2}.$$

For even orders p, q, define

$$D_{p,q}(w) = \sum_{n=0}^{\infty} A_p(n) A_q(n) x_n^{-w}, \qquad \operatorname{Re} w > 3 - p - q, \qquad D_r = D_{p,q}(-2r)$$

by meromorphic continuation at the last arguments. The source proves regularity there. Put

$$C_j = \sum_{r=0}^{j} \binom{2j+1}{2r} 2^{-2(j-r)} D_r.$$

Its inverse transform is

$$D_r = \sum_{j=0}^{r} \frac{\binom{2r}{2j}}{2j+1} B_{2r-2j}(1/2) C_j.$$

For $p = 2a$, $q = 2b$, $a < b$, the source shows that all late C_j, $j \geq b$, have rational coordinates on $Z_k = \zeta(2k)$, $0 \leq k < b$, where $Z_0 = \zeta(0) = -1/2$. Thus the formal coordinate space used here has the $2b$ basis labels

$$C_0, \ldots, C_{b-1}, \quad Z_0, \ldots, Z_{b-1}.$$

Their specified numerical values define an evaluation map from this formal rational vector space; the labels are independent by definition, and the numerical values are not asserted to be independent. The exact inverse Bernoulli transforms, late-coordinate interpretation, and omitted-binomial convention are inherited from [7, 6]. The coordinate matrix of $C_0, \ldots, C_{2b-1}$ is $\operatorname{diag}(I_b, M_{a,b})$. The displayed inverse transform from C to D has diagonal entries $1/(2r+1)$, which explains the full-prefix determinant factor.

Let $p = 2a$, $q = 2b$, $1 \leq a < b$, and put

$$\mu_j = \begin{cases} B_{2j}/(2j)!, & j \geq 0, \\ 0, & j < 0. \end{cases}$$

We use $B_1 = -1/2$; in particular $\mu_0 = 1$, and the zero values at negative indices are a convention, not analytic continuation. The source matrix for the late coordinates of the first q centered moments is

$$(M_{a,b})_{tk} = \mu_{t+k-a+1} \left\{ \frac{(2b+2t-2a+1)!}{(2b-2k)!} + \mathbf{1}_{k<a} \frac{(2t+1)!}{(2a-2k)!} \right\}, \qquad 0 \leq t, k < b.$$

Its full prefix determinant is $\det M_{a,b}/(4b-1)!!$. Let $H_{a,b} = (\mu_{t+k-a+1})$ and let U be the rank-a perturbation in the source factorization $M_{a,b} = R(H_{a,b} + U)E$. Define

$$\rho_{a,b} = \frac{\det(H_{a,b} + U)}{\det H_{a,b}}.$$

The source proves $H_{a,b}$ nonsingular [7]; the diagonal factors R, E are nonzero. Consequently $\rho_{a,b} \neq 0$ is equivalent to the optimal first-q formal-rank assertion. These are rational matrix identities, not assertions of arithmetic independence of zeta values.

For the finite contraction let $0 \leq i, \ell < a$ and set

$$n_i = b - a + i, \qquad N_{i\ell} = 2(n_i - \ell) + 1, \qquad R_\ell = 2b - 2a + 2 + 2\ell,$$

$$F_{i\ell} = f_\ell(n_i), \qquad f_\ell(n) = \frac{1}{(2\ell+1)!} \prod_{r=0}^{2\ell-1} (2n - r)(2n + 3 + r).$$

For $W_{i\ell}(z) = (z + 2\ell + 1)(2n_i + 2\ell + z + 1)^{\underline{2i}}$, put

$$d_{i\ell h} = \frac{1}{h!}\sum_{j=0}^{h}(-1)^{h-j}\binom{h}{j}W_{i\ell}(N_{i\ell} - j).$$

Using independent symbols $U_1, \dots, U_{2a-1}$ define

$$\mathcal{A}^{[a]}_{i\ell} = 2F_{i\ell} + \frac{R_\ell!}{(2n_i + 1)(2n_i + 2)(2\ell + 2)!N_{i\ell}!}\sum_{\substack{1\le h\le 2i+1\\ h \text{ odd}}} d_{i\ell h}N_{i\ell}^{\underline{h}}U_{a-i+\ell+(h-1)/2}. \tag{7.1}$$

Factorial ratios are finite rational functions of b, since $R_\ell - N_{i\ell} = 1 + 4\ell - 2i$ is fixed. For $b \ge 2a$ the source small-determinant theorem gives

$$\rho_{a,b} = \left.\frac{\det \mathcal{A}^{[a]}}{\det F}\right|_{U_j = 2B_{2b-2j}}. \tag{7.2}$$

Here is the normalization check, including the endpoint correction. The source Padé row transformation, followed by reversal of the first a columns and division of row i by c_{n_i}, takes the unperturbed entries to $F_{i\ell}$. Expand $W_{i\ell}(z) = \sum_h d_{i\ell h}(N_{i\ell} - z)^{\underline{h}}$ in its finite Newton basis and use

$$\sum_{j=0}^{N}\binom{N}{j}B_j(N - j)^{\underline{h}} = N^{\underline{h}}B_{N-h}(1).$$

For $b \ge 2a$, $N - h \ge 2$ for retained odd h, and the even h terms have odd index at least three and vanish. Restoring the omitted $j = 1$ term contributes precisely a second $F_{i\ell}$. This gives (7.1); the factors two in $2F$ and $U_j = 2B_{2b-2j}$ are both necessary. At smaller b, the exceptional B_0 and $B_1(1)$ terms require the direct finite calculation.

7.2 The new five-square integer certificate

Put

$$\begin{aligned} D_{10}(b) = {} & 4096b^{12} - 110592b^{11} + 1236992b^{10} - 7303680b^9 \\ & + 23705088b^8 - 38140416b^7 + 14925056b^6 + 24490080b^5 \\ & - 4469984b^4 - 15838992b^3 - 2571948b^2 + 4244400b + 1488375. \end{aligned}$$

For $a = 5$, direct finite polynomial identities give

$$\begin{aligned} \det F = {} & \frac{16384}{4465125}(b - 3)(b - 2)^2(b - 1)(2b - 7)(2b - 5)^2(2b - 3) \\ & \cdot (4b - 15)(4b - 13)(4b - 11)^2(4b - 9)^2(4b - 7)^2(4b - 5)(4b - 3)D_{10}(b), \end{aligned} \tag{7.3}$$

$$\frac{\det \mathcal{A}^{[5]}}{\det F} = \frac{P_{10}(b; U_1, \dots, U_9)}{138240(2b - 7)(2b - 5)D_{10}(b)}, \tag{7.4}$$

where $P_{10} \in \mathbb{Z}[b, U_1, \dots, U_9]$. The polynomial has 439 monomials in the nine U variables; their coefficients have degree at most 24 in b. In the full ten-variable expansion there are 10,403 monomials. Its complete coefficients are supplied with the exact verification artifact. Equivalently, (7.1), (7.3), and (7.4) specify it without interpolation or an omitted fitted formula.

The identities can be verified by clearing the least common denominator of each row of (7.1), summing the 120 signed permutation products, and cross-multiplying (7.4). All coefficient comparisons are over the rational polynomial ring. The delivered verifier performs these operations exactly and checks the integer coefficients of P_{10}. Since $D_{10}(b)$ is odd at every integer b, the denominator in (7.4) always has valuation 10.

For order eight we retain the current source polynomial [6], normalized by

$$\rho_{4,b} = \frac{P_8(b; 2B_{2b-2}, \dots, 2B_{2b-14})}{576(2b-5)D_8(b)}, \qquad b \geq 8,$$

where

$$D_8(b) = 256b^8 - 3584b^7 + 18496b^6 - 40544b^5 + 25984b^4 + 18424b^3 - 5856b^2 - 8316b - 2025.$$

Its denominator has valuation 6. The independently regenerated P_8 agrees coefficientwise with the source certificate.

7.3 A uniform Bernoulli congruence at prescribed precision

All congruences are in $\mathbb{Z}_{(2)}$, the rationals with odd denominator. We use the source's elementary fact $v_2(B_{2j}) = -1$ for $j \geq 1$; all Bernoulli numbers have valuation at least -1.

Lemma 7.1. *Let $K \geq 4$ and $m \geq 1$ be integers, let $s = \lceil K/2 \rceil$, and suppose $2m \geq K + s - 1$. Set $J = \lfloor (K + s - 2)/3 \rfloor$. Then*

$$2B_{2m} \equiv \frac{1}{2^{s-1}} \sum_{\substack{1 \leq a < 2^s \\ a \text{ odd}}} \sum_{j=0}^{J} \binom{m}{j} (a^2 - 1)^j \pmod{2^K}.$$

The right side is a finite integer-valued polynomial in m.

Proof. Faulhaber's identity gives

$$\sum_{a=0}^{2^s-1} a^{2m} = 2^s B_{2m} + \sum_{h=2}^{2m+1} \binom{2m}{h-1} \frac{2^{sh}}{h} B_{2m+1-h}.$$

The $h = 2$ term is zero. For $h \geq 3$, its valuation is at least $sh - v_2(h) - 1 \geq s + K - 1$. Thus division by 2^{s-1} proves $2B_{2m} \equiv 2^{1-s} \sum a^{2m} \pmod{2^K}$. The even summands vanish at this precision because $2m \geq K + s - 1$. For odd a, $a^2 - 1$ is divisible by eight. Expanding $a^{2m} = (1 + (a^2 - 1))^m$ and discarding $j > J$ gives an error of valuation at least $3j - s + 1 \geq K$ in each term. This proves the formula. For completeness, sums of even powers of the odd integers below 2^s are divisible by 2^{s-1}: pair a with $a + 2^{s-1}$ and induct on s. Thus the coefficients in the binomial basis are integers. The finite average may also be retained in its displayed rational form; no claim about dividing a congruence by a nonunit without increasing its precision is being made. □

For $K = 16$, $s = 8$, finite coefficient simplification gives the particularly useful integer polynomial

$$\begin{aligned} \mathcal{B}(x) &= 1 + 20068x - 26384x^2 + 2816x^3 - 19712x^4 + 4096x^5 - 24576x^6, \\ 2B_{2m} &\equiv \mathcal{B}(m) \pmod{65536}, \qquad m \geq 12. \end{aligned} \tag{7.5}$$

For an entirely finite specification of this simplification, expand

$$\frac{1}{128} \sum_{a=1,3,\dots,255} \sum_{j=0}^{7} \binom{x}{j} (a^2 - 1)^j.$$

Every coefficient of its difference from $\mathcal{B}(x)$ has valuation at least 16. The exact program checks this in $\mathbb{Q}[x]$, including the denominators from $\binom{x}{j}$. The degree-seven coefficient vanishes at the displayed precision.

7.4 Two short congruences and their unique exceptional classes

Define

$$\begin{aligned} R_8(b) &= 4096b^9 - 5120b^8 + 4864b^7 + 2176b^6 - 176b^5 \\ &\quad - 5184b^4 - 4912b^3 - 7948b^2 + 1195b - 1727, \\ R_{10}(b) &= -512b^8 + 128b^7 - 928b^6 - 816b^5 + 464b^4 + 352b^3 + 118b^2 + 217b + 694. \end{aligned}$$

The following are coefficientwise polynomial congruences:

$$P_8(b; \mathcal{B}(b-1), \dots, \mathcal{B}(b-7)) \equiv 4R_8(b) \pmod{65536}, \tag{7.6}$$
$$P_{10}(b; \mathcal{B}(b-1), \dots, \mathcal{B}(b-9)) \equiv 32R_{10}(b) \pmod{65536}. \tag{7.7}$$

They follow by direct integer polynomial substitution in the finite certificates; the accompanying program performs exact modular polynomial multiplication and verifies every coefficient. After the actual Bernoulli coordinates are substituted, the precision after division is explicitly

$$\frac{P_8(b; 2B_{2b-2}, \dots, 2B_{2b-14})}{4} \equiv R_8(b) \pmod{16384} \qquad (b \geq 19),$$
$$\frac{P_{10}(b; 2B_{2b-2}, \dots, 2B_{2b-18})}{32} \equiv R_{10}(b) \pmod{2048} \qquad (b \geq 21).$$

All coefficients of degree at least two in either R_8 or R_{10} are even, and the linear coefficients are odd. Consequently, for all integers x, y,

$$v_2(R_j(x) - R_j(y)) = v_2(x - y), \qquad x \neq y, \quad j = 8, 10. \tag{7.8}$$

Indeed, after factoring $x - y$, the remaining polynomial is the odd linear coefficient plus an even integer. In particular, a root modulo 2^L is unique if it exists. Exact substitution gives

$$R_8(8001) \equiv 0 \pmod{16384}, \qquad R_{10}(962) \equiv 0 \pmod{2048}.$$

Thus these are the unique exceptional classes at their respective precisions. Their existence and uniqueness do not rely on testing all residues.

Theorem 7.2 (Order-eight refinement)**.** *For every integer $b \geq 5$ with $b \not\equiv 8001 \pmod{16384}$, $M_{4,b}$ is nonsingular. Equivalently, the first $q = 2b$ centered moments have full formal rank q for $(p, q) = (8, 2b)$. For $b \geq 19$ in this range,*

$$v_2(\rho_{4,b}) = v_2(b - 8001) - 4.$$

Proof. For $b \geq 19$, each $b - j \geq 12$ in the seven Bernoulli coordinates. Apply (7.5) and (7.6). If b is not in the exceptional class, (7.8) gives $v_2(R_8(b)) = v_2(b - 8001) < 14$. Hence the true numerator has valuation $2 + v_2(b - 8001)$, and division by the denominator of valuation 6 proves the assertion. The existing order-eight theorem handles $5 \leq b \leq 18$ except $b = 17$. Direct calculation gives

$$\rho_{4,17} = -\frac{5564518155020169678178976040991}{8937041637533625} \neq 0.$$

□

Theorem 7.3 (Order-ten prefix)**.** *For every integer $b \geq 6$ with $b \not\equiv 962 \pmod{2048}$, $M_{5,b}$ is nonsingular. Equivalently, the first $q = 2b$ centered moments have full formal rank q for $(p, q) = (10, 2b)$. For $b \geq 21$ in this range,*

$$v_2(\rho_{5,b}) = v_2(b - 962) - 5.$$

Proof. For $b \geq 21$, all nine indices $b-j$ are at least 12. Equations (7.5), (7.7), and (7.8) give $v_2(P_{10}) = 5 + v_2(b-962)$ whenever $b \not\equiv 962 \pmod{2048}$. The denominator in (7.4) has valuation 10. This proves nonvanishing and the valuation formula. For $6 \leq b \leq 20$, an exact finite calculation with the displayed source matrix gives the respective valuations

$$-4, -4, -6, -5, -2, -5, -4, -5, -3, -5, -4, -5, -1, -5, -4.$$

The complete nonzero determinants and ratios are recorded in `direct_rank_checks.json`; its independent verifier constructs $M_{5,b}$ and $H_{5,b}$ from Bernoulli numbers and takes their exact rational determinants. It does not invoke the generic contraction below its valid threshold $b \geq 10$. □

For example, the previously unsettled gap-six pair $(p,q) = (10,16)$ is included among these finite base cases, with

$$\det M_{5,8} = \frac{537653622905129}{68669669376000} > 0.$$

The first pair in the excluded order-ten progression has $b = 962$; its exclusion is a limitation of the congruence, not evidence of a zero determinant. The same applies to $b = 8001$ in the order-eight refinement. The global gap-six conjecture remains open here.

7.5 A useful lower precision corollary and exact scope

The same finite calculation at modulus 512 gives

$$P_{10}(b; \mathcal{B}_9(b-j)) \equiv 32(6b^2 - 7b + 6) \pmod{512},$$

where $\mathcal{B}_9(x) = 256x^4 + 256x^3 + 240x^2 + 100x + 1$ represents $2B_{2m}$ modulo 512 for $m \geq 7$. The odd linear coefficient again gives a unique zero, now $b \equiv 2 \pmod{16}$. Thus a shorter statement is that order ten is proved outside $b \equiv 2 \pmod{16}$, with $v_2(\rho_{5,b}) = v_2(b-2) - 5$ for $b \geq 16$ outside that class. The higher precision theorem treats many additional values within it.

Further increases in precision are algorithmic through Lemma 7.1, but an arbitrarily long list of certified residue classes is not a proof for all b. In particular, neither the order-eight nor the order-ten formula above asserts a global valuation law obtained by replacing the exceptional class with a single ordinary integer. A proof for the remaining progressions, or an independent archimedean nonvanishing argument, is still needed.

8 Source reconciliation and further research

8.1 Corrections, qualifications, and status changes

Current source status. The targeted audit did not establish an additional false theorem in the current source dependencies. The useful corrections are therefore precise integration and scope corrections, not a claim to have found a new error merely because a stronger statement is unavailable. The two Stieltjes corrections already incorporated in the unified manuscript must be retained. Its chapter `11-stieltjes-source-corrections.tex` explicitly repairs an initial-index discrepancy in the inspected version of a harmonic-zeta paper and a literal-coefficient sign description in a height-one multiple-zeta paper [1]. Neither is a new discovery of this delivery.

The exact conventions can be checked without a numerical fit. From (1.1) and the Hurwitz recurrence,

$$\gamma_r(a+1) = \gamma_r(a) - \frac{\log^r a}{a}.$$

Consequently a Stieltjes limit whose sum begins at $n = 1$ instead of $n = 0$ has lost $\log^r(a)/a$. For $r = 0, a = 1$ the lost quantity is one. Likewise the literal coefficient of u in $\zeta(1+u, a)$ is $-\gamma_1(a)$, rather than $\gamma_1(a)$. These facts explain both inherited corrections and fix the sign convention needed for (2.10) and (4.16).

Higher-order pole terms. For a product $C_k(s)\zeta(1+2s,a)$, the coefficient of s^r uses the coefficient of s^{r+1} in C_k. Formula (4.16) retains this term as $r!\alpha_{k,r+1}/2$. A proposed extension obtained by differentiating only the regular Laurent part would be incorrect. The present article supplies the complete replacement, without suggesting that the source's proved first-derivative theorem was wrong.

Ordinary sums and continued values. The continuous-order square law is an ordinary identity only on its stated half-plane. At $s = 0$ and a fixed positive a, continuation gives $\mathcal{B}_0(z) = 1/2$. The continued lower-end expression in (2.8) is then $1/4$, although every pointwise summand of its left side at $s = 0$ is zero. In the finite identity (2.9), the upper endpoint is exactly $1/4$ and cancels it. This is a concrete obstruction to silently extending the ordinary summation prescription across that boundary. Similarly, the finitely many low-order coefficients in (6.11) are continued spectral values when the unmodified weighted series diverges. The article states that distinction explicitly.

Moment reconstruction. The gap-four theorem is already proved in [4, 3]. An older research-status paragraph that still lists it as wholly open should be updated by citing those reports. For order eight, the inherited theorem outside $b \equiv 1 \pmod{16}$ is strengthened here; it is not relabelled as an original result. The new statements are the smaller exceptional class in Theorem 7.2 and the order-ten family in Theorem 7.3.

Experimental valuation laws must be replaced by their proved finite-precision versions. For example, the new order-eight formula gives valuations 4 at $b = 65$ and 2 at $b = 129$, while the order-ten formula gives valuation 2 at $b = 66$. These follow from $b - 8001$ and $b - 962$, respectively. They do not justify a global law centered at an ordinary integer: the excluded progressions remain outside the theorem. Moreover, formal full rank refers to the specified moment/zeta coordinate module. It must not be converted into arithmetic independence of its numerical evaluations.

Cyclotomic parity. The finite coefficient identity in Section 3 is a form of the classical colored depth-two parity theorem [9]. The addition is its explicit rational-tail resummation with arbitrary outer character, its cutoff proof, and the specialized evaluations. The complementary conjugation sector is not reduced. Conditional bilateral intermediate sums are locally uniformly convergent; the projected double sums used in the final generator are absolutely normally convergent. Replacing the first statement by absolute normal convergence would be an unjustified strengthening.

Gaussian conjectures and formal obstructions. The source's Gaussian weight-six and revised weight-eight conjectures remain unresolved here. A non-membership result in a specified shuffle, parity, or Cayley relation space is a limitation of that relation system. It is not a disproof of an equality between actual periods. The newly evaluated Gaussian rational tails have their own definitions and cannot be substituted for those older conjectural quantities. No integration note should mark the older problems solved on the strength of the present conductor-four examples.

8.2 A proposed research programme with concrete endpoints

Each question below asks for an identity, a complete structural description, or an exact nonvanishing certificate. The order reflects dependence on the proved results, not a claim about difficulty.

Question 1 (The two remaining moment progressions)**.** Can one prove nonvanishing in the excluded classes $b \equiv 8001 \pmod{16384}$ for order eight and $b \equiv 962 \pmod{2048}$ for order ten? Lemma 7.1 makes higher congruence precision effective. A useful next result would either control the exceptional classes uniformly at every precision or give an independent exact argument on them. Certifying any fixed number of additional binary digits still leaves an infinite progression.

One must also distinguish a compatible 2-adic root from an ordinary positive integer at which the rational determinant actually vanishes.

Question 2 (The full gap-six family)**.** For every even $p = 2a$ and $q = 2a+6$, does the first-q prefix have full formal rank? The finite example $(p, q) = (10, 16)$ is included here, but a proof uniform in a is still missing. The fixed-order contraction has the threshold $b \geq 2a$, whereas here $b = a + 3$. For $a > 3$ its omitted B_0 and B_1 contributions cannot be discarded. An exact bordered-Hankel formula that retains those moving endpoint terms is a concrete next target.

Question 3 (All unequal even orders)**.** Find a uniform nonvanishing theorem for $M_{a,b}$, $1 \leq a < b$, or identify and prove the exact exceptional set if one exists. A useful intermediate result would characterize the small determinant after the Padé transformation in terms of explicit orthogonal polynomial evaluations and their finite differences. The endpoint should be an exact identity or sign/nonzero certificate, rather than an extrapolation from a large table of ranks.

Question 4 (Bilinear adjacent-tail certificates)**.** Replace E_n^2 in Theorem 2.1 by a product of tails with distinct polynomial denominators P, Q. Allow both products of adjacent tails in the certificate. Which rational coefficient functions eliminate every linear-tail term, and which pairs P, Q admit a finite endpoint evaluation? The compatibility calculation in Section 2 classifies only the displayed one-product ansatz. The first nontrivial test is two distinct quadratic resolvents with independent shifts.

Question 5 (Continuation of the ordinary square law)**.** Continue (2.8) across $\Re s = 0$ with a fully specified subtraction of its exact upper endpoint. Determine the contact term and all order derivatives at each nonpositive integer. The $s = 0$ discrepancy of $1/4$ described above is a mandatory first test. An all-index formula would connect negative-order alternating Hurwitz values, Euler polynomials, and Laurent coefficients without confusing ordinary sums with continued values.

Question 6 (Independent shifts and independent order jets)**.** The two factors of $P(k) = (k+a)(k+b)$ in (2.6) belong to one tail. They do not answer the source question about $\zeta(p, n+v)\zeta(q, n+w)$ with independently shifted tails. Derive a finite-cutoff identity for those products, then differentiate independently in p, q, and the outer weight order. The desired formula must state all moving polar hyperplanes and their residues. This would extend the matched continuous-order diagonal treated here and resolve the corresponding independent-shift question in [3].

Question 7 (The complementary cyclotomic sector)**.** Construct the two-resolvent generator for

$$\frac{R_{a,b;c}(x, y) - R_{a^{-1},b^{-1};c^{-1}}(x, y)}{2i}$$

in the odd-weight setting. A first conductor-four coefficient is $\Im Z(2, 1; i, -i)$. Specify a candidate collection of classical or cyclotomic polylogarithms and an exact reduction into it, or retain a precisely defined additional coordinate. The parity generator alone does not determine this sector. Failure of a finite numerical search would not prove independence from the proposed collection.

Question 8 (Shifted and degree-generating cyclotomic normal forms)**.** Introduce independent rational shifts into the bilateral lattice proof and compute its finite boundary terms. Then introduce a third generating variable that sums the outer polynomial degree h in (3.13). Having a formula for every individual h does not yet produce that analytic generating function. A compact answer would organize all resolvent and weight derivatives by a finite Bell-polynomial rule, with its convergence region stated.

Question 9 (Three-tail products)**.** Develop a three-tail analogue of the finite outer-cutoff decomposition. The natural target is an explicit depth-two remainder plus finite single-polylogarithm terms and endpoints. Determine the exact parity sector in which this reduction occurs. The product of three tails has a new ordered region structure, so unrestricted depth-one closure would be an unsupported extrapolation from Section 3.

Question 10 (A finite description of the logarithmic Gamma correction)**.** At which rational shifts and nonzero masses can $S(a, b)$ in (5.1) be expressed in a stated finite collection of polylogarithm values and order derivatives? The elementary-kernel forms (5.5)–(5.6) and the exact shift/distribution rules provide starting equations. A rigorous finite evaluation at a single nontrivial mass would be progress. For generic parameters, the Gamma integral remains a complete exact representation; no minimality theorem for that representation has been established.

Question 11 (Several masses at every spectral order)**.** Extend Theorem 6.1 to $r \geq 3$. Pointwise, the variance identity is replaced by a higher symmetric polynomial in the logarithms L_j. Determine the finite-subtracted kernels, their confluent coefficients, and their mixed mass primitives. With even weights there is no pole at zero; with odd weights, compute the additional Stieltjes and one-higher-coefficient residue terms rather than assuming the even-weight prescription still applies.

Question 12 (Shift, mass, and order derivatives in one formula)**.** Combine the primitive tower J_m with the divided-difference kernel $R_{P,a}$. Can every rectangular mixed shift–mass primitive of the second defect be written using a finite, explicitly anchored collection of Hurwitz derivatives and ordinary integrals? The mixed derivatives must commute on a stated domain, and constants must be fixed at all base faces. This is a stronger target than the separate one-variable and double-mass primitives proved here.

Question 13 (Joint tail identities and spectral identities)**.** The square certificate, the cyclotomic parity generator, and the quadratic resolvent all produce polygamma expressions. At a fixed conductor and weight, form their joint exact relation system and determine which mixed harmonic-tail coefficients it isolates. A particularly concrete endpoint is a new closed formula for a single mixed tail product, rather than only the convolution in (2.5). Keep the formal relation quotient distinct from the space of numerical periods.

Question 14 (Small trusted certificates and formalization)**.** Which of the finite proofs can be reduced to short rational identities that a proof assistant can check directly? The adjacent-tail recurrence, the harmonic coefficient formula, the resonant Laurent multiplication, and the two modular substitution certificates are natural first targets. The analytic continuation lemmas then need explicit locally uniform majorants and an agreed special-function library. A formal check of the finite data alone would not formalize the associated infinite-dimensional theorem.

9 Reproducibility and integration

9.1 Exact algebra and independent numerical diagnostics

The companion programs use Python, SymPy, and mpmath. They are self-contained relative to the delivered certificate files and require no private paths or network access. The recorded environment uses Python 3.12.14, SymPy 1.14.0, and mpmath 1.3.0. The proofs themselves do not depend on a particular floating-point library.

Component	Exact verification	Numerical diagnostics
Tail square and Stieltjes identities	96 exact assertions, including the finite endpoint and deliberately incorrect color-sign rejection	29 diagnostics at 75 decimal digits; independent integral, finite-cutoff, and order-derivative calculations.
Cyclotomic generator	Written bilateral proof and finite parity coefficient extraction in Section 3	12 diagnostics at 60–65 digits, including trivial colors, arbitrary outer character, odd conductor, rational examples, and the Gamma primitive.
Spectral defects and mass kernels	194 exact assertions for harmonic coefficients, Laurent multiplication, logarithmic kernels, variance, and confluence	21 diagnostics at 65 digits, including independent spectral continuation, Gamma integrals, mass differentiation, and anchored primitives.
Moment-rank refinements	Complete rational polynomial determinant replay, coefficientwise modular substitutions, and original finite source-matrix determinants	No floating rank test is used as evidence for a nonvanishing theorem.

The 62 high-precision checks complement the 290 exact assertions in the two analytic verification suites. The larger determinant certificates are a separate exact calculation, not included in that assertion count. The order-ten numerator has 10,403 integer monomials in its full expansion. The verifier reconstructs the determinant from the displayed rational matrix and cross-multiplies coefficientwise; it does not infer a polynomial by interpolation at sample values.

The numerical spectral comparison does not merely evaluate the same closed form twice. It constructs the quadratic zeta by a separate Pochhammer–Hurwitz expansion, differentiates the finite linear-factor Hurwitz expression, and compares their difference with the finite-subtracted logarithmic formula. The tail checks use finite endpoint identities and independent integral representations where appropriate. Nonetheless, all decimal results remain diagnostics. The recorded error tolerances are not rigorous enclosures of the infinite sums.

The determinant proof was independently reviewed against the original binomial source matrix, in addition to the factorial form used in the delivered verifier. The Bernoulli thresholds, division of moduli, source normalization, and exceptional classes were also checked independently. The exact replay and the written proof have different roles: the replay validates finite algebra; the uniform congruence argument makes that finite algebra apply to infinitely many b.

9.2 Files and commands

The package has two equivalent LaTeX entry points: `article.tex` inputs the modular sections, while `Alternating_Tails_and_Spectral_Defects.tex` contains the complete standalone article. The supplied PDF is compiled from the standalone source. The modular files make thematic

integration straightforward. The bibliography is local to the package; no external bibliography database is required.

From the extracted package directory, the verification entry point is

```
python verification/run_all.py
```

The exact determinant replay may take longer than the small symbolic checks. A diagnostic shortcut, which intentionally omits reconstruction of the two multivariate determinant identities, is

```
python verification/verify_rank_refinement.py --quick
```

Quick mode is useful during editing; the recorded delivery was checked with the full replay. The programs rewrite their result JSON files. For an independent review that preserves the delivered evidence, run them in a copy of the extracted package.

The PDF can be rebuilt with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
    Alternating_Tails_and_Spectral_Defects.tex
```

The package includes a short build helper that first regenerates the standalone source from the modular files. The only nonstandard Python dependencies are pinned in `requirements.txt`. No source repository checkout is required to reproduce the new certificates.

9.3 Suggested placement and claim ledger

This continuation belongs with the harmonic-tail and spectral/Stieltjes thematic reports under `Analysis/Polylogarithms/docs/reports/`. The delivered directory can be preserved intact and cross-referenced from both themes. Later editorial integration into the unified manuscript can use Sections 2–3 for tail identities, Sections 4–6 for spectral derivatives and primitives, and Section 7 for moment rank. The current incoming procedure treats placement and manuscript rewriting as separate stages [2].

The machine-readable claim ledger records what is proved, inherited, partially advanced, and still open. In particular it keeps the Gaussian conjectures, complementary cyclotomic sector, independent tail shifts, global gap-six family, and two exceptional moment progressions open. The provenance record names the exact repository pin and the six inspected archive sources. The integration notes identify the source research questions updated by the new theorems and preserve the earlier correction history.

References

[1] V. Reshetnikov, *ProveIt: Polylogarithms unified manuscript*, repository snapshot `73f18ef013d5`, 11 October 2026. Pinned manuscript directory.

[2] *ProveIt incoming reports*, six archives at repository snapshot `73f18ef013d5`, 11 October 2026. Pinned incoming directory.

[3] *Harmonic Tails, Shifts, and Twists*, research continuation for the ProveIt programme, 11 October 2026. Delivered source `ProveIt_Harmonic_Tails_Shifts_and_Twists_2026-10-11.zip`, in [2]; especially the twisted-tail generator and Q6.

[4] *Harmonic-Tail and Gamma Continuation*, research continuation for the ProveIt programme, 11 October 2026. Delivered source `ProveIt_Harmonic_Tail_Gamma_Continuation_2026-10-11.zip`, in [2]; gap-four moment-prefix theorem and source status.

[5] *Quadratic-Spectrum Stieltjes Identities: Complete reflected jets, harmonic counterterms, finite polylogarithmic reductions, and a polynomial parity criterion*, research continuation for the ProveIt programme, 10 October 2026. Delivered source `ProveIt_Quadratic_Spectrum_Stieltjes_2026-10-10.zip`, in [2]; further-research questions 2 and 7.

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[15] NIST Digital Library of Mathematical Functions, Sections 5.4–5.5, *Gamma function: special values and functional relations*; in particular Gauss's rational digamma formula (5.4.19), reflection, and multiplication. Accessed 11 October 2026. https://dlmf.nist.gov/5.4; https://dlmf.nist.gov/5.5.